Numerical analysis of superconducting phases in the extended Hubbard model with non-local pairing

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Abstract

[To be continued...]

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List of symbols and abbreviations

AF	Antiferromagnetic
BCS	Bardeen-Cooper-Schrieffer (theory)
β	Inverse temperature
$\Delta^{(\gamma)}$	γ-wave component of the gap function
DoF	Degree(s) of freedom
EHM	Extended Hubbard model
EV(s)	Expectation value(s)
FS	Fermi surface
HF	Hartree-Fock
HFP(s)	Hartree-Fock parameter(s)
HM	Hubbard model
L_ℓ	Number of lattice sites in direction ℓ
LRT	Linear Response Theory
MFT	Mean-Field Theory
NN(s)	Nearest neighbor(s)
RWC(s)	Relevant Wick's contraction(s)
RMP(s)	Renormalized model parameter(s)
SC	Superconductor
SDW	Spin-density wave
SSB	Spontaneous symmetry breaking
T_c	Critical temperature

Chapter 1

Mean-field theory discussion of the EHM

This chapter is devoted to develop a Mean Field Theory (MFT) framework of the Extended Hubbard model (EHM) of Eq. (**??**),

$$\hat{H} = \underbrace{-t\sum_{\langle ij\rangle}\sum_{\sigma}\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}}_{\hat{H}_t} \underbrace{+U\sum_{i}\hat{n}_{i\uparrow}\hat{n}_{i\downarrow}}_{\hat{H}_U} \underbrace{-V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{n}_{i\sigma}\hat{n}_{j\sigma'}}_{\hat{H}_V}$$

Mean Field Theory (MFT) is a widely used and simple theoretical tool, often sufficient to describe the leading orders in phase transition phenomena of Many-Body Physics. Here MFT is employed generically, in order to later discuss both the effects of the non-local term $\hat{H}_V$ onto the AF phase (Chap. 3), as well as the insurgence of anisotropic superconductivity (Chap. **??**). The latter, following the path of Bardeen-Cooper-Schrieffer (BCS) theory in describing conventional s-wave superconductivity. As will be thoroughly described, the lattice spatial structure directly influences the topology of the gap function, giving rise to anisotropic pairing.

[Move this part to Chap. **??**] Sec. **??** studies the non-local attraction $\hat{H}_V$ in real-space, describing how such interaction can contribute to the hamiltonian as a symmetry-breaking term in given channels. In the following sections, we move to specific channels and study theoretically and numerically the effect of non-local attraction.

1.1 Extended Hubbard model symmetries and Wick's theorem

The general aim of this project is to study the phase diagram of the EHM by comparing ground-state energies of different phases. The phases we consider in next chapters for the EHM are the normal state (N), the antiferromagnetic ordering (AF), given by a non-uniform distribution of charge in each spin sector, and the (anisotropic) superconducting phase, described by a uniformly distributed charge allowing for Cooper pairing instabilities. The first evident symmetry of the model is a global U(2) symmetry [**arovas2022hubbard**] given by the transformation

$$\hat{c}_{i\sigma} \to \sum_{\sigma'}\mathcal{U}_{\sigma\sigma'}\hat{c}_{i\sigma'} \quad \text{with} \quad \mathcal{U}_{\sigma\sigma'} \in \mathrm{U}(2) \tag{1.1}$$

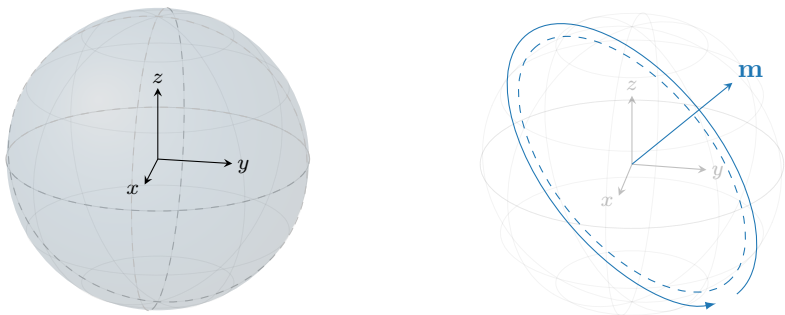
The proof is trivial and follows from the fact that $\mathcal{U}^{\dagger}\mathcal{U} = \mathcal{U}\mathcal{U}^{\dagger} = \mathbb{I}$. Note that the the following decomposition holds,

$$\mathrm{U}(2) = \mathrm{SU}(2) \otimes \mathrm{U}(1) \tag{1.2}$$

which expresses the separate charge conservation and spin rotational symmetries. Thus, for the EHM, three symmetries are "brekable":

1. Discrete translational invariance. By breaking explicitly this symmetry, the obtained state must show a Charge-Density Wave (CDW) ordering;

2. $\mathrm{U}^{\mathrm{c}}(1)$ charge conservation, coming from Eq. (1.2) decomposition. By breaking this symmetry, we allow for the total charge to fluctuate in the ground state;



a | SU(2) rotations domain before SSB.

b | U(1) rotations domain after SSB.

Figure 1.1 | Graphic representation of the concept of SU(2) → U(1) SSB. The blue color identifies, in both cases, the rotations domain for the current symmetry. When SSB takes place in a magnetic phase, the magnetization order parameter acquires a precise direction which was previously invariant over the full spherical domain. After SSB, the hamiltonian is rotationally invariant only for rotations *around* the selected magnetization vector.

Symmetry group	**Operations**
Point group	Discrete translations on lattice
$\mathrm{U}^{\mathrm{c}}(1)$	Charge-conserving global phase shifts
SU(2)	Three-dimensional rotations in spin space
$\mathrm{U}^{z}(1)$	Reduced two-dimensional rotations around order parameter after SSB

Table 1.1 | Model symmetries and relative group operations. Note that $\mathrm{U}^{\mathrm{c}}(1)$ represents the charge conserving group, while $\mathrm{U}^{z}(1)$ represents the subgroup obtained by breaking the SU(2) symmetry with a vector order parameter and preserving its amplitude.

3. SU(2) spin rotation symmetry, also , coming from Eq. (1.2) decomposition. If this symmetry is broken, we allow for the ground state to exhibit a preferred spin direction.

4. Note that the SU(2) group contains itself a U(1) subgroup. SU(2) symmetry can be broken, selecting a particular vectorial direction for the order parameter, and reduced to a smaller $\mathrm{U}^{z}(1)$ symmetry which is essentially expressed by the conservation of the *magnitude* of the order parameter.

The last point is important: for the AF phase, for example, there is no need for the magnetization vector to be directed along the *real* z-axis, which is the one perpendicular to the lattice plane. SU(2) symmetry breaks when (staggered) magnetization is established along a particular direction, but spin rotations around said direction still are ground state symmetries. The symmetries are reported synthetically in Tab. 1.1.

Different phases are described by different order parameters, each one exhibiting a specific symmetry subset from the initial set, while the rest are broken. Ferromagnetic state perform SU(2) → $\mathrm{U}^{z}(1)$ spontaneous symmetry breaking (SSB), through the magnetic order parameter acquiring a specific spatial direction. Such SSB process is graphically depicted in Fig. 1.1. Antiferromagnetic ordering breaks also translational invariance (reducing it to a smaller symmetry, relative to double-sized unitary cells). Basic superconducting models do not break translational invariance and can preserve full SU(2) symmetry, while breaking $\mathrm{U}^{\mathrm{c}}(1)$ charge conservation. The three phases hereby considered do not completely break translational invariance: the latter either is left untouched or is reduced to a weaker translational invariance. Because of this, it will be rather useful to move the framework to reciprocal space. We will get there in Sec. **??**.

1.1.1 Particle-hole symmetry at half-filling

[To be continued...]

1.1.2 The role of spin

[To be continued...]

1.1.3 Translational invariance

[To be continued...]

1.2 MFT on Hubbard lattices

[To be continued...]

1.2.1 MFT general procedure and self-consistency

The Mean Field Theory approach to a fermionic quartic is given by the following procedure: given the hamiltonian made of the quadratic operators $\hat{A}$ and $\hat{B}$

$$\hat{H} = \hat{A}\hat{B}$$

we define the fluctuations

$$\delta\hat{A} = \hat{A} - \langle\hat{A}\rangle \qquad \delta\hat{B} = \hat{B} - \langle\hat{B}\rangle$$

thus the following chain holds:

$$\begin{aligned}\hat{A}\hat{B} &= \langle\hat{A}\rangle\langle\hat{B}\rangle + \delta\hat{A}\langle\hat{B}\rangle + \langle\hat{A}\rangle\delta\hat{B} + \delta\hat{A}\delta\hat{B} \\ &= -\langle\hat{A}\rangle\langle\hat{B}\rangle + \hat{A}\langle\hat{B}\rangle + \langle\hat{A}\rangle\hat{B} + \underbrace{\delta\hat{A}\delta\hat{B}}_{\mathcal{O}(\delta)^2}\end{aligned}$$

which implies

$$\langle\hat{H}\rangle = \langle\hat{A}\rangle\langle\hat{B}\rangle + \langle\delta\hat{A}\delta\hat{B}\rangle$$

Thus, if $|\langle\hat{A}\rangle\langle\hat{B}\rangle| \gg |\langle\delta\hat{A}\delta\hat{B}\rangle|$, in a statistical sense we can neglect the $\mathcal{O}(\delta)^2$ part of the hamiltonian and approximate

$$\hat{H} \simeq \hat{A}\langle\hat{B}\rangle + \langle\hat{A}\rangle\hat{B} - \langle\hat{A}\rangle\langle\hat{B}\rangle$$

Self consistency of MFT procedure is thus embodied by the condition

$$\left|\frac{\langle\delta\hat{A}\delta\hat{B}\rangle}{\langle\hat{A}\rangle\langle\hat{B}\rangle}\right| \ll 1 \qquad \Longrightarrow \qquad \left|\frac{\langle\hat{A}\hat{B}\rangle}{\langle\hat{A}\rangle\langle\hat{B}\rangle} - 1\right| \ll 1 \tag{1.3}$$

which essentially states that the least the fluctuations, the better the approximation. This rule of thumb is necessary in order to interpret the physical relevance of numerical MFT results. We will come back to this in the end, for now assuming the validity of MFT approximation and deriving the mathematical results.

1.2.2 MFT treatment of the pure Hubbard model

The pure Hubbard model, $\hat{H}_t + \hat{H}_U$, offers a simple framework for MFT analysis of the AF phase: this is described in detail in App. **??**. The key passage is there given by the MFT approximation

$$\hat{n}_{i\uparrow}\hat{n}_{i\downarrow} \simeq \hat{n}_{i\uparrow}\langle\hat{n}_{i\downarrow}\rangle + \langle\hat{n}_{i\uparrow}\rangle\hat{n}_{i\downarrow} + (\text{constants}) \tag{1.4}$$

from which the AF structure is simply recovered. In terms of the discussion of the above paragraph, we have $\hat{A} \equiv \hat{n}_{i\uparrow}$ and $\hat{B} \simeq \hat{n}_{i\downarrow}$. However the "full" operator is given by

$$\hat{n}_{i\uparrow}\hat{n}_{i\downarrow} = \hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{j\downarrow}\hat{c}_{j\downarrow}\hat{c}_{i\uparrow}$$

Term	Normal phase	Antiferromagnet	Superconductor
Hartree			
Fock	Prohibited	Prohibited	Prohibited
Cooper	Prohibited	Prohibited	

Table 1.2 | Summary of Wick's terms selection rules and relevant contributions to the effective hamiltonian for each phase limitedly to $\hat{H}_U$.

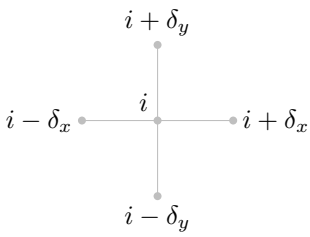


Figure 1.2 | Schematic representation of the four NNs of a given site i for a planar square lattice.

which contracts in three different ways following Wick's Theorem:

$$\langle \hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{i\downarrow}\hat{c}_{i\downarrow}\hat{c}_{i\uparrow}\rangle \simeq \underbrace{\langle \hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{i\downarrow}\rangle\langle \hat{c}_{i\downarrow}\hat{c}_{i\uparrow}\rangle}_{\text{Cooper}} - \underbrace{\langle \hat{c}^\dagger_{i\uparrow}\hat{c}_{i\downarrow}\rangle\langle \hat{c}^\dagger_{i\downarrow}\hat{c}_{i\uparrow}\rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_{i\uparrow}\hat{c}_{i\uparrow}\rangle\langle \hat{c}^\dagger_{i\downarrow}\hat{c}_{i\downarrow}\rangle}_{\text{Hartree}} \tag{1.5}$$

As a first approximation, the theorem is assumed to hold (which, in a BCS-like fashion, is equivalent to assuming for the ground-state to be a coherent state). The AF ground state breaks translational invariance and reduces rotational symmetry, $\mathrm{SU}(2) \to \mathrm{U}^z(1)$. Thus, of the three terms above, when considering the pure Hubbard model in AF phase:

- Cooper fluctuations are suppressed, because they break charge conservation;
- Similarly the Fock term is null as well because if $i = j$ and $\sigma' = \overline{\sigma}$ (as is for the local interaction, which contains the operator $\hat{n}_{i\uparrow}\hat{n}_{i\downarrow}$) the expectation values involved are describing a process breaking the survivor $\mathrm{U}^z(1)$ spin symmetry.

Thus, correctly, the Wick's decomposition of Eq. (1.4) only involves Hartree-terms of Eq. (1.5). The distinction between the normal phase and the AF phase id given by whether the Hartree fluctuations break or not translational symmetry: antiferromagnetism is indeed a specific SDW ordering which weakens the latter. In general the three parts of the decomposition (Hartree, Fock, Cooper) need to be considered altogether: this is what is done in the next chapters. What said here extends naturally to the EHM: the relevant contributions for each phase given by the local repulsion $\hat{U}$ in the EHM are listed in Tab. 1.2.

1.3 The non-local attraction $\hat{H}_V$ as a source of symmetry-breaking interactions

Let us now extend the discussion to the EHM and consider the NN non-local attraction,

$$\hat{H}_V \equiv -V \sum_{\langle ij\rangle} \sum_{\sigma\sigma'} \hat{n}_{i\sigma}\hat{n}_{j\sigma'} \tag{1.6}$$

Evidently the hamiltonian can be decomposed in various spin terms,

$$\begin{aligned} \hat{H}_V &= \sum_{\sigma\sigma'} \hat{H}_V^{\sigma\sigma'} \\ &= \underbrace{\hat{H}_V^{\uparrow\uparrow} + \hat{H}_V^{\downarrow\downarrow}}_{\text{Same-spin}} + \underbrace{\hat{H}_V^{\uparrow\downarrow} + \hat{H}_V^{\downarrow\uparrow}}_{\text{Opposite-spin}} \end{aligned} \tag{1.7}$$

The role of said terms is crucial in establishing the pairing channel of the dominant physical phase. As an example, the s.s. terms can be seen as a possible source of triplet pairing superconductivity. Next section are devoted to analyze said terms and derive a simple analytical result in reciprocal space.

1.3.1 Real space description

Let us start from the real space form of Eq. (1.7). To carry out a summation over nearest neighbors $\langle ij \rangle$ of a square lattice means precisely to sum over all links of the lattice. Then we can identify the generic opposite-spin (o.s.) term $\hat{H}_V^{\sigma\overline{\sigma}}$ as the one collecting the σ operators of sublattice $\mathcal{S}_a$ and $\overline{\sigma}$ operators of sublattice $\mathcal{S}_b$. The o.s. non-local interactions can be written as a sum of terms over just one of the two sublattices $\mathcal{S}_a$ and $\mathcal{S}_b$, oppositely polarized in the AF configuration (see Fig. **??**). Define the site hamiltonian

$$\hat{h}_V^{(i)} = -V \sum_{\ell=x,y} (\hat{n}_{i\uparrow}\hat{n}_{i+\delta_\ell\downarrow} + \hat{n}_{i\uparrow}\hat{n}_{i-\delta_\ell\downarrow}) \tag{1.8}$$

Then, it follows,

$$\begin{aligned} \hat{H}_V^{(\text{o.s.})} &= \overbrace{\sum_{i\in\mathcal{S}_a} \hat{h}_V^{(i)}}^{\hat{H}_V^{\uparrow\downarrow}} + \overbrace{\sum_{i\in\mathcal{S}_b} \hat{h}_V^{(i)}}^{\hat{H}_V^{\downarrow\uparrow}} \\ &= \sum_{i\in\mathcal{S}} \hat{h}_V^{(i)} \end{aligned} \tag{1.9}$$

Here the notation of Fig. **??** is used. The two-dimensional lattice is regular-square. For each site i in a given sublattice, the nearest neighbors sites are four – all in the other sublattice. The notation used is $i \pm \delta_x$, $i \pm \delta_y$ as in Fig. 1.2. Similarly, the same-spin (s.s.) hamiltonian decomposes as

$$\hat{H}_V^{(\text{s.s.})} = -V \sum_{i\in\mathcal{S}_a} \sum_{\ell=x,y} \sum_{\sigma} (\hat{n}_{i\sigma}\hat{n}_{i+\delta_\ell\sigma} + \hat{n}_{i\sigma}\hat{n}_{i-\delta_\ell\sigma})$$

Note here the summation only on one sublattice. The non-local interaction contribution to energy, as a function of the $T = 0$ full hamiltonian ground-state[1] $|\Psi\rangle$, is given by

$$\begin{aligned} E_V[\Psi] &= \langle\Psi|\hat{H}_V|\Psi\rangle \\ &= -V \sum_{\langle ij\rangle} \sum_{\sigma\sigma'} \langle \hat{n}_{i\sigma}\hat{n}_{j\sigma'} \rangle \\ &= \underbrace{-V \sum_{\langle ij\rangle} \sum_{\sigma} \langle \hat{n}_{i\sigma}\hat{n}_{j\sigma} \rangle}_{\text{s.s.}} \underbrace{-V \sum_{\langle ij\rangle} \sum_{\sigma} \langle \hat{n}_{i\sigma}\hat{n}_{j\overline{\sigma}} \rangle}_{\text{o.s.}} \end{aligned} \tag{1.10}$$

Shorthand notation has been used: $\langle\Psi|\cdot|\Psi\rangle = \langle\cdot\rangle$. The ground-state must realize the condition

$$\frac{\delta}{\delta\langle\Psi|} E[\Psi] = 0$$

being $E[\Psi]$ the total energy (made up of the three terms of couplings t, U and V).

[Expand derivation.]

The functional derivative must be carried out in a variational fashion including a Lagrange multiplier, the latter accounting for state-norm conservation, as is done normally in deriving the Hartree-Fock approximation for the eigenenergies of the electron liquid [**grosso2014solid**, **giuliani2005quantum**].

[1] Extensions to finite temperatures is simple: minimization must be carried out on free energy, while expectation values must be taken in a thermodynamic fashion.

Sector	Term	Normal phase	Antiferromagnet	Superconductor
o.s.	Hartree			
	Fock	Prohibited	Prohibited	Prohibited
	Cooper	Prohibited	Prohibited	
s.s.	Hartree			
	Fock			
	Cooper	Prohibited	Prohibited	

Table 1.3 Summary of Wick's terms selection rules and relevant contributions to the effective hamiltonian for each phase limitedly to $\hat{H}_V$.

Opposite-spin terms. Consider first the o.s. terms of Eq. (1.10): take e.g. the term $\hat{n}_{i\uparrow}\hat{n}_{i+\delta_x\downarrow}$. As in Eq. (1.5), Wick's Theorem states that, if the expectation value is performed onto a coherent state,

$$\begin{aligned}\langle \hat{n}_{i\uparrow}\hat{n}_{i+\delta_x\downarrow}\rangle &= \langle \hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{i+\delta_x\downarrow}\hat{c}_{i+\delta_x\downarrow}\hat{c}_{i\uparrow}\rangle \\ &= \underbrace{\langle \hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{i+\delta_x\downarrow}\rangle\langle \hat{c}_{i+\delta_x\downarrow}\hat{c}_{i\uparrow}\rangle}_{\text{Cooper}} - \underbrace{\langle \hat{c}^\dagger_{i\uparrow}\hat{c}_{i+\delta_x\downarrow}\rangle\langle \hat{c}^\dagger_{i+\delta_x\downarrow}\hat{c}_{i\uparrow}\rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_{i\uparrow}\hat{c}_{i\uparrow}\rangle\langle \hat{c}^\dagger_{i+\delta_x\downarrow}\hat{c}_{i+\delta_x\downarrow}\rangle}_{\text{Hartree}}\end{aligned}$$

Identical decompositions are given for all others NNs. Of the three terms above:

- The Coooper term breaks $\mathrm{U}^{\mathrm{c}}(1)$ charge symmetry, allowing for supeconducting instabilities;
- The Fock term breaks the $\mathrm{U}^z(1)$ symmetry, because it accounts for a site hop *plus* spin flip process;
- The Hartree term can break translational invariance (based on our Ansatz on the expectation value), because the mean-field to interact with is given by the local density. SU(2) symmetry is also broken, because we do not have spin DoF perfect degeneracy anymore, but $\mathrm{U}^z(1)$ symmetry still holds.

As a general rule of thumb, throughout this text we will discard the o.s. Fock term eveywhere. For the normal state as well as for the AF instability only the Hartree term is to be accounted; in superconducting instability also the Cooper term contributes. Note that for superconducting instabilities, due to superexchange mechanism (as explained in App. **??**) the o.s. term account for singlet pairing as well as zero-spin triplet pairing. Which channel is preferred, is a matter of thermodynamic advantage.

Same-spin terms. Consider then the same-spin terms: take e.g. the term $\hat{n}_{i\uparrow}\hat{n}_{i+\delta_x\uparrow}$. As above,

$$\begin{aligned}\langle \hat{n}_{i\uparrow}\hat{n}_{i+\delta_x\uparrow}\rangle &= \langle \hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{i+\delta_x\uparrow}\hat{c}_{i+\delta_x\uparrow}\hat{c}_{i\uparrow}\rangle \\ &= \underbrace{\langle \hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{i+\delta_x\uparrow}\rangle\langle \hat{c}_{i+\delta_x\uparrow}\hat{c}_{i\uparrow}\rangle}_{\text{Cooper}} - \underbrace{\langle \hat{c}^\dagger_{i\uparrow}\hat{c}_{i+\delta_x\uparrow}\rangle\langle \hat{c}^\dagger_{i+\delta_x\uparrow}\hat{c}_{i\uparrow}\rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_{i\uparrow}\hat{c}_{i\uparrow}\rangle\langle \hat{c}^\dagger_{i+\delta_x\uparrow}\hat{c}_{i+\delta_x\uparrow}\rangle}_{\text{Hartree}}\end{aligned}$$

Identical consideration as in the above paragraph hold for each term. The only difference with the o.s. terms is given by the Fock term: since the spin-flip process is absent, now the Fock fluctuations actually contribute as an effective NN hopping term. In other words, this term does not break $\mathrm{U}^z(1)$ symmetry and thus is perfectly legitimate in each phgase. Tab. 1.3 summarizes the selection rules hereby presented.

As a final remark, notice that the superconducting instabilities of the s.s. terms account only for triplet pairing. The only possible superconducting ordering established by the means of these terms is odd in real space. Then s-wave and d-wave superconductivity cannot establish in this channel; p_ℓ-wave superconductivity, instead, can.

1.3.2 Reciprocal space transform of the model interactions

It is useful to derive analytically the reciprocal-space form of both the U and V interactions. Let us start from the latter.

Non-local attraction. Consider a generic bond, say, the one connecting sites j and $j \pm \delta_\ell$ (variable i is here referred to as the imaginary unit to avoid confusion). $\mathbf{x}_j$ is the 2D notation for the position of site j, while $\boldsymbol{\delta}_\ell$ is the 2D notation for the lattice spacing previously indicated as δ_ℓ. Fourier transform it according to the convention

$$\hat{c}_{j\sigma} = \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}\in \mathrm{BZ}} e^{-i\mathbf{k}\cdot\mathbf{x}_j} \hat{c}_{\mathbf{k}\sigma}$$

Then:

$$\begin{aligned}\hat{n}_{j\sigma}\hat{n}_{j\pm\delta_\ell\sigma'} &= \hat{c}^\dagger_{j\sigma}\hat{c}^\dagger_{j\pm\delta_\ell\sigma'}\hat{c}_{j\pm\delta_\ell\sigma'}\hat{c}_{j\sigma} \\ &= \frac{1}{(L_xL_y)^2}\sum_{\nu=1}^{4}\sum_{\mathbf{k}_\nu\in\mathrm{BZ}} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{x}_j} e^{\pm i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}_\ell}\hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma}\end{aligned}$$

Then, the interaction at site j, spin σ with its NNs at spin σ' – indicated as $(j\sigma\sigma')$ – is given by

$$\begin{aligned}(j\sigma\sigma') &= -V\sum_{\ell=x,y}\sum_{\delta=\pm\delta_\ell}\hat{n}_{j\sigma}\hat{n}_{j\pm\delta_\ell\sigma'} \\ &= -\frac{V}{(L_xL_y)^2}\sum_{\ell=x,y}\sum_{\nu=1}^{4}\sum_{\mathbf{k}_\nu\in\mathrm{BZ}} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{x}_j} \\ &\qquad\qquad \times\left(e^{i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}_\ell}+e^{-i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}_\ell}\right)\hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma} \\ &= -\frac{2V}{(L_xL_y)^2}\sum_{\ell=x,y}\sum_{\nu=1}^{4}\sum_{\mathbf{k}_\nu\in\mathrm{BZ}} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{x}_j}\cos\left[(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}_\ell\right]\hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma}\end{aligned}$$

The full non-local interaction is given by summing over all sites of $\mathcal{S}_a$ (which is, half the sites of $\mathcal{S}$). This gives back momentum conservation,

$$\frac{1}{L_xL_y}\sum_{j\in\mathcal{S}_a} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{x}_j} = \frac{1}{2}\delta_{\mathbf{k}_1+\mathbf{k}_2=\mathbf{k}_3+\mathbf{k}_4}$$

Let $\mathbf{k}_1+\mathbf{k}_2=\mathbf{k}_3+\mathbf{k}_4=\mathbf{K}$, and define $\mathbf{k}$, $\mathbf{k}'$ such that

$$\mathbf{k}_1\equiv\mathbf{K}+\mathbf{k} \qquad \mathbf{k}_2\equiv\mathbf{K}-\mathbf{k} \qquad \mathbf{k}_3\equiv\mathbf{K}-\mathbf{k}' \qquad \mathbf{k}_4\equiv\mathbf{K}+\mathbf{k}' \qquad \delta\mathbf{k}\equiv\mathbf{k}-\mathbf{k}'$$

Sums over these variables must be intended as over the Brillouin Zone (BZ). Then, finally

$$\begin{aligned}\hat{H}_V &= \sum_{j\in\mathcal{S}_a}\sum_{\sigma\sigma'}(j\sigma\sigma') \\ &= -\frac{V}{L_xL_y}\sum_{\sigma\sigma'}\sum_{\ell=x,y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\cos\left(\delta\mathbf{k}\cdot\boldsymbol{\delta}_\ell\right)\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma'}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \\ &= -\frac{V}{L_xL_y}\sum_{\sigma\sigma'}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma'}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \qquad (1.11)\end{aligned}$$

The result here obtained will be used various times in next chapters. Note that different Wick contraction schemes lead to different results:

$$\overbracket{c^\dagger_{\mathbf{K}+\mathbf{k}\sigma}c^\dagger_{\mathbf{K}-\mathbf{k}\sigma'}}c_{\mathbf{K}-\mathbf{k}'\sigma'}c_{\mathbf{K}+\mathbf{k}'\sigma} \qquad \text{Cooper contraction} \qquad (1.12)$$

$$\overbracket{c^\dagger_{\mathbf{K}+\mathbf{k}\sigma}c^\dagger_{\mathbf{K}-\mathbf{k}\sigma'}c_{\mathbf{K}-\mathbf{k}'\sigma'}}c_{\mathbf{K}+\mathbf{k}'\sigma} \qquad \text{Fock contraction} \qquad (1.13)$$

$$\overbracket{c^\dagger_{\mathbf{K}+\mathbf{k}\sigma}c^\dagger_{\mathbf{K}-\mathbf{k}\sigma'}c_{\mathbf{K}-\mathbf{k}'\sigma'}c_{\mathbf{K}+\mathbf{k}'\sigma}} \qquad \text{Hartree contraction} \qquad (1.14)$$

which will be used explicitly later on.

Local repulsion. A somewhat identical argument can be carried out for the local interaction,

$$\hat{H}_U = U \sum_{i \in \mathcal{S}} \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}$$

The only difference here is made by the fact that the interaction is repulsive on-site, thus the final structure factor in the Fourier transform disappears as well as the minus sign,

$$\hat{H}_U = \frac{U}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \tag{1.15}$$

When contracting these fermionic operators, identical considerations as for the non-local counterpart (1.11) hold.

1.4 General computational approach to MFT

[To be continued...]

1.4.1 Sketch of the Hartree-Fock iterative algorithm

The general sketch of a self-consistent Hartree-Fock algorithm in our context is simple enough. Start from a fermionic quartic hamiltonian,

$$\hat{H} = \sum_{\alpha\beta} A_{\alpha\beta} \hat{c}^\dagger_\alpha \hat{c}_\beta + \sum_{\alpha\beta\gamma\delta} B_{\alpha\beta\gamma\delta} \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta + \text{h.c.} \tag{1.16}$$

where $A_{\alpha\beta}$, $B_{\alpha\beta\gamma\delta}$ are model parameters. We limit ourselves to quartic hamiltonians, but the idea can be extended. As discussed before, in this kind of context MFT is employed by applying Wick's theorem. This way, the quartic hamiltonian is effectively reduced to a quadratic hamiltonian,

$$\hat{H} \simeq \hat{H}^{(\text{eff})} = \sum_{\alpha\beta} C_{\alpha\beta} \hat{c}^\dagger_\alpha \hat{c}_\beta + \sum_{\alpha\beta} D_{\alpha\beta} \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta + \text{h.c.} \tag{1.17}$$

where now the new parameters, which we will refer to as Hartree-Fock parameters (HFP), depend on the model itself via the equations

$$C_{\alpha\beta} = A_{\alpha\beta} + \sum_{\gamma\delta} \left[B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta} \right] \left\langle \hat{c}^\dagger_\gamma \hat{c}_\delta \right\rangle$$

$$D_{\alpha\beta} = \sum_{\gamma\delta} B_{\alpha\beta\gamma\delta} \left\langle \hat{c}_\gamma \hat{c}_\delta \right\rangle$$

These equations are determined by many-body expectation values, which in general are computed by the means of thermal averages over the lowest energy density matrix. A self consistent Hartree-Fock scheme goes as follows:

1. Apply MFT to the quartic hamiltonian, recovering its general theoretic effective form;
2. Impose selection rules on the parameters $C_{\alpha\beta}$, $D_{\alpha\beta}$, eventually eliminating those not respecting the symmetry sector we are working in[2];
3. Choose a starting value and a coherent symmetry structure for the remaining parameters;
4. Enter iterative scheme of Fig. 1.3:
   a) Based on the current HFPs, determine $\mu[C, D]$ that gives the correct density;
   b) Diagonalize the effective hamiltonian $\hat{H}^{(\text{eff})}[C, D] - \mu[C, D]\hat{N}$ and recover the ground-state $|\Psi[C, D]\rangle$;

[2] As an example: if we choose not to break charge conservation, set $\forall \alpha, \beta : \; D_{\alpha\beta} = 0$.

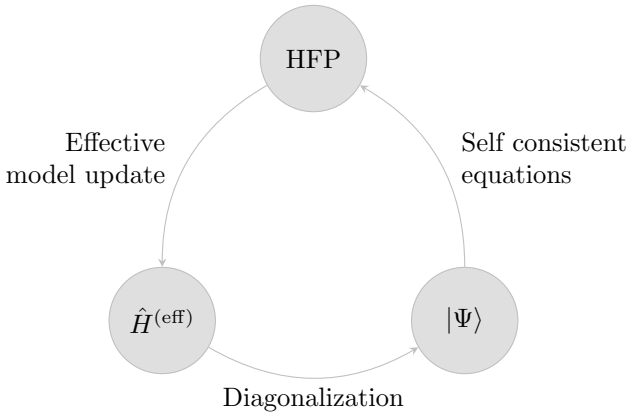


Figure 1.3 Schematization of the self-consistent Hartree Fock method described in Sec. 1.4. The effective model is solved, the solution is used to estimate self-consistently the Hartree Fock vector, which is then used to update iteratively the model itself. [Update including initializer and endpoint.]

c) Compute the new HFPs $C'_{\alpha\beta}$, $D'_{\alpha\beta}$ via the self consistent equations;

d) Initialize the new effective hamiltonian $\hat{H}^{(\mathrm{eff})}[C', D']$ and repeat the cycle until the HFPs have converged to stable values.

Of course, to work any of these algorithms we need to know how to diagonalize the hamiltonian. In the context of the Extended Hubbad Model at thermal equilibrium, due to the phases we are simulating, we know analytically the diagonalized hamiltonian, and making use of this knowledge we are able to reduce the self consistent scheme to a simple iterative algorithm. An explicit example of HF algorithm used in present text, somewhat simplified due to the presence of a single HFP, is given in App. **??**.

1.4.2 Self-consistency and RMPs

As will be thoroughly discussed in the following derivations for each physical phase we treat, treating the EHM via MFT keeping under strict control the effects of Wick's theorem produces ground states whose parametric properties (band structures, gap functions, chemical potential) are controlled parametrically and self-consistently by the model two-body interactions $\hat{H}_U$ and $\hat{H}_V$. Let us define the concept of Renormalized model parameter (RMP), meaning one model that is effectively renormalized via interactions.

Let us invent a relevant example to explain the concept of RMP. Suppose that under MFT, for a given phase the decomposition $\hat{H}_V$ has a net effect of defining the RMP $\tilde{t}$,

$$t \xrightarrow{V} \tilde{t}$$

In other words, the decomposition of $\hat{H}_V$ produces a free Fermionic hamiltonian whose diffusive parameter is $\tilde{t}$. We define $\tilde{t}$ as an RMP, being the result of the original model parameter t plus the effect of interactions.

1.4.3 Algorithm parametrization

[To be continued...]

1.4.4 Fixed density simulations and the role of μ

In the context of the present text, all simulations are performed at fixed density, determining self-consistently the value of the chemical potential. This means that, for a given step i and the

relative HFPs vector $\mathbf{v}_i$, and for a fixed density $n \in [0,1]$, μ_i (the chemical potential at the current step) is determined as

$$\mu_i \quad : \quad \min_{\mu} \left\{ \frac{1}{2L_x L_y} \sum_{\mathbf{k}} \sum_{\sigma} \langle \hat{n}_{\mathbf{k}\sigma} \rangle \big|_{\mathbf{v}=\mathbf{v}_i} - n \right\} \tag{1.18}$$

being $\mathbf{v} = \mathbf{v}_i$ and indication that the averaging operation is performed over the ground state determined parametrically by the current value of the HFPs.

Hence, if we are able to perform averages and know both $\mathbf{v}_i$ and n, at step i we infer the correct chemical potential. Now, as we will discuss $\tilde{\mu}$ will turn out to be an RMP itself, meaning that theoretically

$$\mu \to \mu + F[\mathbf{v}] \equiv \tilde{\mu}[\mathbf{v}]$$

being F some functional of the HFPs. In other words, a common effect of interactions under MFT will be precisely to shift the chemical potential. At step i, keeping the density fixed, the chemical potential we are determining numerically and using in our equations is then $\tilde{\mu}_i$. In simpler words, the designed algorithm "does not know" about chemical potential renormalization and thus just determines the number that solves Eq. (1.18). But we know that this number is made of two contributions: the "physical" μ_i and the effective shift produced by the model itself, $F[\mathbf{v}_i]$. So, when it comes to μ_i, this consideration becomes important.

It is the case of the ground state energy estimation. Let us abstract from the EHM and MFT, and consider a generic hamiltonian $\hat{H}$. To find the ground state $|\Psi\rangle$ means precisely to solve

$$\frac{\delta}{\delta \langle \Psi |} \langle \Psi | \hat{H} | \Psi \rangle = 0$$

Now, working with a fixed particles number N means we are performing a constrained minimization (which is, taking the functional derivative) on the composite operator

$$\hat{H} - \mu(\hat{N} - N) \equiv \hat{K} + \mu N$$

being μ the Lagrange multiplier, $\hat{N}$ the number operator and $\hat{K} \equiv \hat{H} - \mu\hat{N}$. Finding the constrained ground state parametrized by a vector $\mathbf{v}$ now means to solve

$$\mathbf{v} \implies |\Psi[\mathbf{v}]\rangle \quad : \quad \begin{cases} \dfrac{\delta}{\delta \langle \Psi |} \langle \Psi | \hat{K} + \mu N | \Psi \rangle \Big|_{|\Psi\rangle = |\Psi[\mathbf{v}]\rangle} = 0 \\ \langle \Psi[\mathbf{v}] | \hat{N} | \Psi[\mathbf{v}] \rangle = N \end{cases}$$

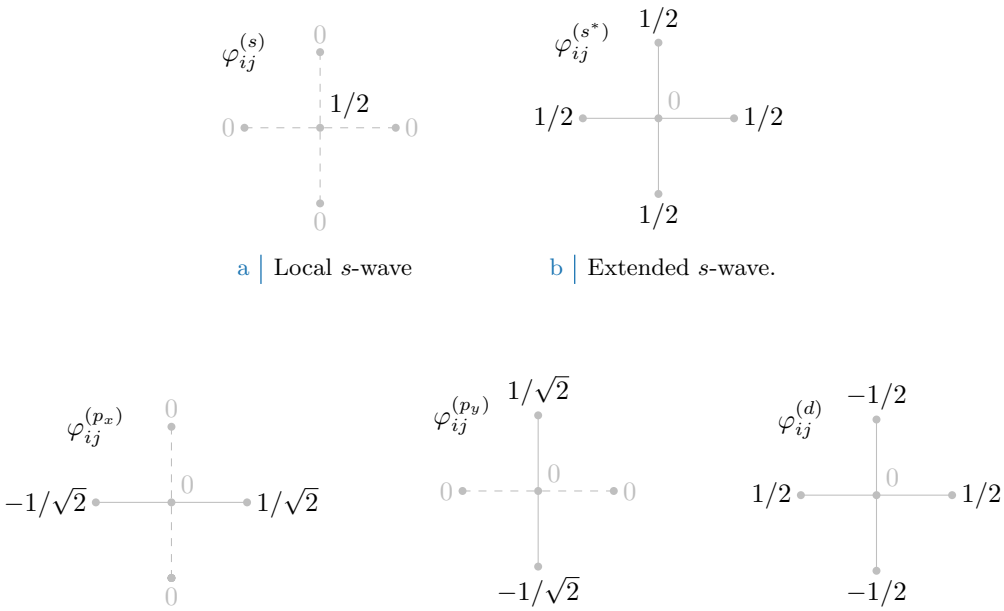
Of course, in the first line μN is derived to zero. The constraint is imposed in the second line. This means that, when it comes to determining the parametric ground state, $\hat{K}$ is functionally minimized. If in this context μ (the Lagrange multiplier) sums with other effective terms arisen from algebraic manipulations of $\hat{H}$ giving a grand total of $\tilde{\mu}$, the number operator $\hat{N}$ average is performed over the many-body ground state $|\Psi[\mathbf{v}]\rangle$ with the total effective chemical potential $\tilde{\mu}$, not μ. However, when it comes to computing energy,

$$\begin{aligned} \langle \hat{H} \rangle &= \langle \Psi[\mathbf{v}] | \hat{K} | \Psi[\mathbf{v}] \rangle + \mu N \\ &= \Omega[\mathbf{v}] + (\tilde{\mu}[\mathbf{v}] - F[\mathbf{v}])N \end{aligned}$$

being Ω the expectation value (EV) of $\hat{K}$. In the last passage we wrote everything in terms of functionals of the HFPs vector $\mathbf{v}$. Using this relation, when the algorithm has converged, we can estimate the physical energy.

1.5 The role of 2D symmetry channels

Looking at $\hat{H}$, the main difference between its two-body interactions $\hat{H}_U$ and $\hat{H}_V$ of course relies in locality (first) and non-locality (second). As any two-body interaction, $\hat{H}_V$ can be decomposed in harmonics. At a first look one may think that, being the potential symmetric by $\pi/2$ rotations on the lattice, the resulting ground state should exhibit only properties with identical symmetry. However this is not the case: said symmetry is a property of the hamiltonian operator on the square lattice, however the resulting ground state is a composite object living in a many-body Fock space, [To be continued...]



a | Local s-wave b | Extended s-wave.

c | p_x-wave. d | p_y-wave. e | $d_{x^2-y^2}$-wave.

Figure 1.4 | Form factors at different topologies, as listed in Tab. 1.4. In figures five sites are represented: the hub site i and its four NN. Solid lines represent non-zero values for $\varphi_{\boldsymbol{\delta}}$, while dashed lines represent vanishing factors.

Structure	Form factor	Graph
s-wave	$\varphi_{ij}^{(s)} = \delta_{ij}$	Fig. 1.4a
Extended s-wave	$\varphi_{ij}^{(s^*)} = \frac{1}{2}\left[\delta_{j=i+\delta_x} + \delta_{j=i-\delta_x} + \delta_{j=i+\delta_y} + \delta_{j=i-\delta_y}\right]$	Fig. 1.4b
p_x-wave	$\varphi_{ij}^{(p_x)} = \frac{1}{\sqrt{2}}\left[\delta_{j=i+\delta_x} - \delta_{j=i-\delta_x}\right]$	Fig. 1.4c
p_y-wave	$\varphi_{ij}^{(p_y)} = \frac{1}{\sqrt{2}}\left[\delta_{j=i+\delta_y} - \delta_{j=i-\delta_y}\right]$	Fig. 1.4d
$d_{x^2-y^2}$-wave	$\varphi_{ij}^{(d)} = \frac{1}{2}\left[\delta_{j=i+\delta_x} + \delta_{j=i-\delta_x} - \delta_{j=i+\delta_y} - \delta_{j=i-\delta_y}\right]$	Fig. 1.4e

Table 1.4 | Form factors for a function defined on a square lattice, with a prefactor included for normalization. The last column indicates the graph representation of each contribution given in Fig. **??**. Subscript $x^2 - y^2$ is omitted for notational clarity.

1.5.1 Square lattice spatial symmetry structures

For the 2D square lattice, various spatial symmetries are possible. Let i be a specific site, and consider a generic function g_{ij} whose domain is given by the site itself and its four NNs (the domain concides with the points of Fig. 1.2). Then it is immediate to see that the function can be written in the following basis

$$g_{ij} = \sum_{\gamma} g^{(\gamma)} \varphi_{ij}^{(\gamma)}$$

where $g^{(\gamma)}$ are the g_{ij} symmetries-decomposition coefficients while $\varphi_{ij}^{(\gamma)}$ are the form factors listed in Tab. 1.4 and plotted in Fig. 1.5, a simple orthonormal rearrangement of the harmonics basis.

Antiferromagnetic ordering considered in this text exhibits a perfect $s \oplus s^*$ topological ordering. Instead, superconductivity is established with a given symmetry – which means, symmetry breaking

Structure	Structure factor	Graph
s-wave	$\varphi_{\mathbf{k}}^{(s)} = 1$	Fig. 1.4a
Extended s-wave	$\varphi_{\mathbf{k}}^{(s^*)} = \cos k_x + \cos k_y$	Fig. 1.4b
p_x-wave	$\varphi_{\mathbf{k}}^{(p_x)} = i\sqrt{2}\sin k_x$	Fig. 1.4c
p_y-wave	$\varphi_{\mathbf{k}}^{(p_y)} = i\sqrt{2}\sin k_y$	Fig. 1.4d
$d_{x^2-y^2}$-wave	$\varphi_{\mathbf{k}}^{(d)} = \cos k_x - \cos k_y$	Fig. 1.4e

Table 1.5 Structure factors derived from the correlation structures of Tab. 1.4. The functions hereby defined are orthonormal, and are plotted in Fig- 1.5.

in the phase transition proceeds in a specific channel. Conventional BCS superconductivity arises from the only possible spatial structure of the local pairing, s-wave – here appearing as a local term (Fig. 1.4a) and extended on a non-local term (Fig. 1.4b). Cuprates exhibit a tendency towards $d_{x^2-y^2}$ SC, while other materials towards p-wave types – eventually with some chirality, as is the case for $p_x \pm ip_y$ SCs. To establish SC under a certain symmetry γ means that Cooper pairs acquire said symmetry.

1.5.2 Symmetry channels structure factors decomposition

When dealing with phases which respect translational symmetry, or at most reduce it to a weaker symmetry by shrinking the Brillouin Zone, it' useful to work in reciprocal space, as discussed in Sec. **??**. As said, a particular phase can show specific spatial symmetries, and so will the Hartree Fock parameters. The key point to notice is that self-consistent equations involve sums (or integrals) to be computed over fermionic states, which can be heavily simplified by employing said symmetries. Consider the form factors of Tab. 1.4, and their Fourier transforms of Tab. 1.5. They satisfy the relations:

$$\frac{1}{L_x L_y}\sum_{ij}\varphi_{ij}^{(\gamma)}\varphi_{ij}^{(\gamma)*} = \delta_{\gamma\gamma'} \qquad \frac{1}{L_x L_y}\sum_{\mathbf{k}}\varphi_{\mathbf{k}}^{(\gamma)}\varphi_{\mathbf{k}}^{(\gamma)*} = \delta_{\gamma\gamma'}$$

If these symmetries are developed by the HFPs, it's of no use to integrate over the entire BZ, since different regions linked by periodicity of the form factors will lead to redundant calculations.

Finally, a very useful detail is given by the following argument. Consider the equation

$$\cos(\delta k_x) + \cos(\delta k_y) = \cos k_x \cos k_x' + \sin k_x \sin k_x' + \cos k_y \cos k_y' + \sin k_y \sin k_y'$$

For the sake of readability, the notations

$$c_\ell \equiv \cos k_\ell \qquad s_\ell \equiv \sin k_\ell \qquad c_\ell' \equiv \cos k_\ell' \qquad s_\ell' \equiv \sin k_\ell'$$

are used. Group the four terms above,

$$\underbrace{\left(c_x c_x' + c_y c_y'\right)}_{\text{Symmetric}} + \underbrace{\left(s_x s_x' + s_y s_y'\right)}_{\text{Anti-symmetric}} \tag{1.19}$$

The first two exhibit inversion symmetry for both arguments $\mathbf{k}$, $\mathbf{k}'$; the second two exhibit anti-symmetry. Decoupling the symmetric part, we obtain

$$c_x c_x' + c_y c_y' = \frac{1}{2}(c_x + c_y)(c_x' + c_y') + \frac{1}{2}(c_x - c_y)(c_x' - c_y')$$

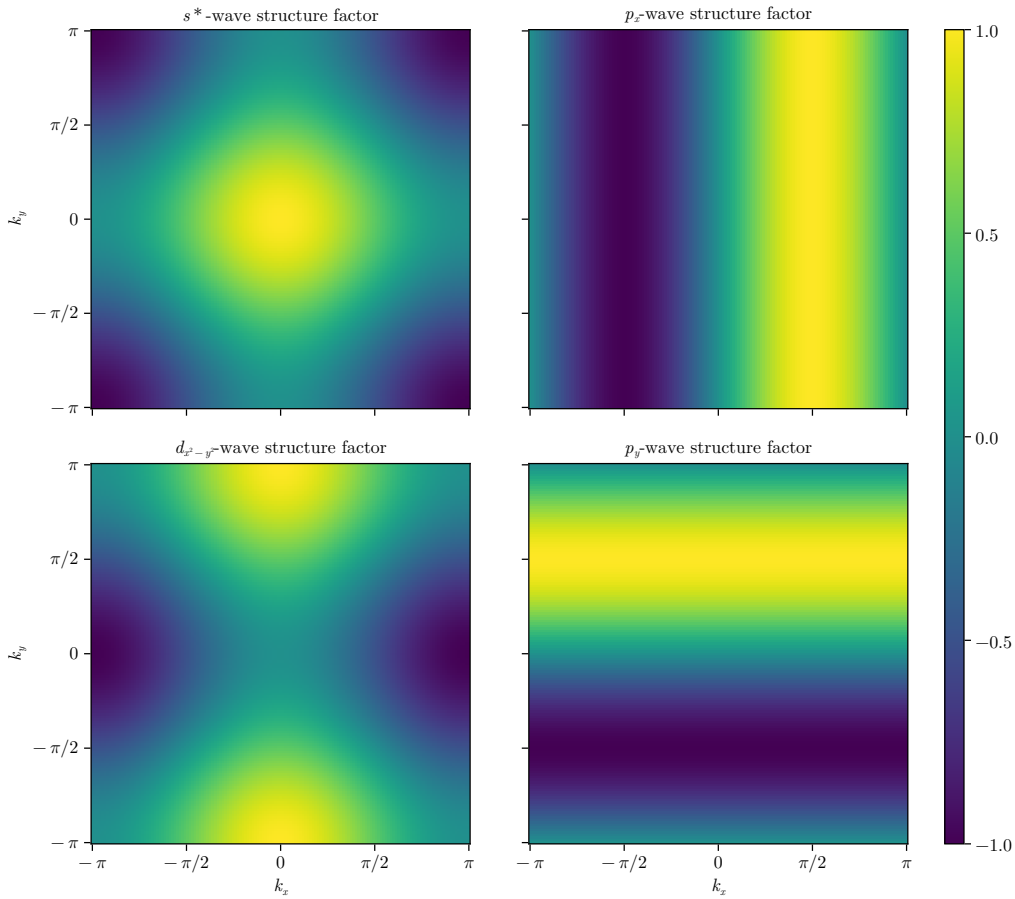


Figure 1.5 | Representation of the various structure factors listed in Tab. 1.5 on the BZ. For the s^* and $d_{x^2-y^2}$-wave factors, the real part is plotted; for the p_ℓ factors, the imaginary part is plotted. [Rotate d-wave]

which finally gives:

$$\begin{aligned}
\cos(\delta k_x) + \cos(\delta k_y) = \frac{1}{2}(c_x + c_y)(c'_x + c'_y) & \qquad (s^*\text{-wave}) \\
+ s_x s'_x & \qquad (p_x\text{-wave}) \\
+ s_y s'_y & \qquad (p_y\text{-wave}) \\
+ \frac{1}{2}(c_x - c_y)(c'_x - c'_y) & \qquad (d_{x^2-y^2}\text{-wave})
\end{aligned}$$

or, in a more compact form

$$\cos(\delta k_x) + \cos(\delta k_y) = \frac{1}{2}\sum_\gamma \varphi^{(\gamma)}_{\mathbf{k}} \varphi^{(\gamma)*}_{\mathbf{k}'} \quad \text{for } \gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\} \tag{1.20}$$

This decomposition will become particularly handy in later calculations.

Chapter 2

The normal phase

This chapter is devoted to the MFT analysis of the "normal phase", which is, the one where interacting electrons do not develop an antiferromagnetic or superconducting gap, although undergoing an effective non-trivial transformation. What will be derived here, in particular the hopping renormalization effect, constitutes a peculiarity of effective MFT treatment of the EHM and will be the groundwork of following analyses. Note that this chapter is structured in such a way that what is derived here mathematically is recovered and expanded to more complex phases in next chapters.

2.1 The "normal" phase and its symmetries

The normal phase is simply the one where no symmetry is broken, and the many-body ground state preserves the entire U(2) structure of Eq. (1.2) as well as translational symmetry. Thus, the normal phase is essentially given by the null gap limit of both the antiferromagnetic and superconducting phases. By discussing it separately now, we highlight a MFT feature of the EHM of particular interest, namely, the insurgence of a new, self-consistent hopping parameter. From Tabs. 1.2 and 1.3, we know that the relevant Wick's contractions (RWCs) for this phase are

$\hat{H}_U$	Hartree contraction
$\hat{H}_V$ (o.s. sector)	Hartree contraction
$\hat{H}_V$ (s.s. sector)	Hartree and Fock contractions

and since the phase we are establishing obeys translational symmetry, we also know Hartree terms are essentially chemical potential shifts. Next section derives this effect in detail.

In the normal phase, the state retains its full $\mathrm{SU}(2) \times \mathrm{U}^{\mathrm{c}}(1) \times$ (Lattice) symmetry discussed in Sec. 1.1. Spin is a pure physical degeneracy, being the $\uparrow$ and $\downarrow$ sectors separated, and $\hat{H}$ is $\mathbb{Z}_2$ invariant with respect to spin inversion, $\sigma \leftrightarrow \overline{\sigma}$. This implies that all physical quantities, as is for the normal phase, should not depend on the spin index. As for the normal phase, we will perform all computations leaving σ explicit and using this consideration at the end.

2.2 Hartree effects: chemical potential renormalization

The first RMP we discuss is μ, whose numerical value is renormalized by Hartree channel interactions within MFT. As is discussed in Sec. 1.4.4, in the context of numerical simulations at fixed density, the chemical potential is determined self-consistently; thus any net effect that shifts μ is effectively ignored within (and limitedly to) the iterative algorithm. Renormalization effects become relevant when computing free energy.

2.2.1 Hartree channel in the normal phase: analytical derivation

Within this section we discuss analytically the effects of MFT on the Hartree channel, separating the local and non-local interactions and obtaining a set of HFPs needed to setup an interative HF algorithm.

Local repulsion. In the normal phase, the local repulsion $\hat{H}_U$ decomposes in Hartree channel as

$$\hat{H}_U \simeq U \sum_i \left[\langle \hat{n}_{i\uparrow}\rangle \hat{n}_{i\downarrow} + \hat{n}_{i\uparrow}\langle \hat{n}_{i\downarrow}\rangle - \langle \hat{n}_{i\downarrow}\rangle\langle \hat{n}_{i\uparrow}\rangle\right]$$

For a translationally invariant phase such as the normal phase, it holds

$$\langle \hat{n}_{i\sigma}\rangle = n \tag{2.1}$$

which gives immediately

$$\begin{aligned}\hat{H}_U &\simeq nU \sum_i \left[\hat{n}_{i\uparrow} + \hat{n}_{i\downarrow}\right] - U \sum_i n^2 \\ &= nU \times \hat{N} - E^{(\mathrm{N})}_{\mathrm{H}/U}\end{aligned}$$

being $\hat{N}$ the total number operator and $E^{(\mathrm{N})}_{\mathrm{H}/U}$ the "contraction energy" (also known as double counting term) due to the Hartree (H) contraction of the U repulsion. The chemical potential is corrected by

$$\mu \to \mu - nU \tag{2.2}$$

The non-local attraction further corrects μ.

Non-local attraction. Let us break down $\hat{H}_V$ isolating its Hartree RWCs in both o.s. and s.s. sectors:

$$\begin{aligned}\hat{H}_V \simeq &\overbrace{-V \sum_{\langle ij\rangle} \sum_\sigma \left[\langle \hat{n}_{i\sigma}\rangle \hat{n}_{j\sigma} + \hat{n}_{i\sigma}\langle \hat{n}_{j\sigma}\rangle - \langle \hat{n}_{i\sigma}\rangle\langle \hat{n}_{j\sigma}\rangle\right]}^{\text{s.s.}} \\ &\underbrace{-V \sum_{\langle ij\rangle} \sum_\sigma \left[\langle \hat{n}_{i\sigma}\rangle \hat{n}_{j\overline{\sigma}} + \hat{n}_{i\sigma}\langle \hat{n}_{j\overline{\sigma}}\rangle - \langle \hat{n}_{i\sigma}\rangle\langle \hat{n}_{j\overline{\sigma}}\rangle\right]}_{\text{o.s.}} + (\text{all the rest})\end{aligned} \tag{2.3}$$

where "all the rest" collects all non-Hartree RWCs. Recalling the Ansatz of Eq. (2.1), we get

$$\hat{H}_V \simeq \underbrace{-nV \sum_{\langle ij\rangle} \sum_\sigma \left[\hat{n}_{j\sigma} + \hat{n}_{i\sigma}\right]}_{\text{s.s.}} \underbrace{-nV \sum_{\langle ij\rangle} \sum_\sigma \left[\hat{n}_{j\overline{\sigma}} + \hat{n}_{i\sigma}\right]}_{\text{o.s.}} - E^{(\mathrm{N})}_{\mathrm{H}/V} + (\text{all the rest})$$

with $E^{(\mathrm{N})}_{\mathrm{H}/V}$ the normal state shift to energy brought by $\hat{H}_V$. Now, evidently both sums above are just a number operator,

$$\hat{H}_V \simeq -2nzV \times \hat{N} - E^{(\mathrm{N})}_{\mathrm{H}/V} + (\text{all the rest})$$

This accounts for the final Hartree shift of μ,

$$\mu \to \mu + 2znV \tag{2.4}$$

Considering the net shift to μ due to both interactions by combining Eqns. (2.2), (2.4), we get

$$\tilde{\mu} \equiv \mu + n(2zV - U) \tag{2.5}$$

This result remains valid throughout all phases discussed in this text: for the AF phase, the SDW character leaves this shift untouched, while the superconducting phase we are discussion is inherently translational invariant.

2.2.2 Hartree contraction energy

The double counting terms accounting for energy shifts in the Hartree channel are, as anticipated:

$$\begin{aligned} E^{(\mathrm{N})}_{\mathrm{H}/U} &= \sum_i \langle \hat{n}_{i\downarrow}\rangle\langle \hat{n}_{i\uparrow}\rangle \\ &= L_x L_y \times n^2 U \end{aligned} \tag{2.6}$$

for the local repulsion and

$$\begin{aligned} E^{(\mathrm{N})}_{\mathrm{H}/V} &= -V \sum_{\langle ij\rangle}\sum_{\sigma} \left[\langle \hat{n}_{i\sigma}\rangle\langle \hat{n}_{j\sigma}\rangle + \langle \hat{n}_{i\sigma}\rangle\langle \hat{n}_{j\overline{\sigma}}\rangle\right] \\ &= -2V \sum_{\langle ij\rangle}\sum_{\sigma} n^2 \\ &= -L_x L_y \times 2zn^2 V \end{aligned} \tag{2.7}$$

for the non-local attraction, being $z = 4$ the coordination number for the 2D square lattice. In the last passage, we used that

$$\begin{aligned} \sum_{\langle ij\rangle}(1) &= \sum_{i\in\mathcal{S}/2}\sum_{j\in\mathrm{NN}(i)}(1) \\ &= \frac{L_x L_y}{2} \times z \end{aligned}$$

as well as spin degeneracy. Note that, correctly, the contraction energy is an extensive quantity. This implies that we can define a contraction energy per site – something we will do later when computing total free energy density.

2.3 Fock effects: bare bands renormalization

Up to now, no relevant changes have been made to the starting situation of the conventional degenerate Fermi gas. The scenario changes now, with the analysis of the Fock RWCs. Even if no gap opens, in the normal phase the particles interact by the means of a non-local attraction, a feature that effectively redefines the bare fermionic bands.

From Wick's decomposition of $\hat{H}_V$, the only allowed Fock term comes from the same-spin part due to SU(2) symmetry selection rules. Let us focus only on this term when decomposing $\hat{H}$:

$$\hat{H}_V \simeq V \sum_{\langle ij\rangle}\sum_{\sigma} \left[\langle \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle \hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\langle \hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle - \langle \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\langle \hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle\right] + (\text{all the rest}) \tag{2.8}$$

(note the + sign in front of the displayed term). A bond-wise hopping amplitude can be defined,

$$\tilde{t}_{ij\sigma} \equiv t - V\langle \hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle$$

The effective diffusive hamiltonian is given by

$$\begin{aligned} \hat{H}_{\tilde{t}} &= \hat{H}_t + V \sum_{\langle ij\rangle}\sum_{\sigma} \left[\langle \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle \hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma} + \text{h.c.}\right] \\ &= -\sum_{\langle ij\rangle}\sum_{\sigma} \left[\tilde{t}_{ij\sigma}\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma} + \text{h.c.}\right] \end{aligned}$$

To proceed, it is useful to move to reciprocal space.

2.3.1 Fock channel in the normal phase: analytical derivation

In reciprocal space, the effective hopping must be transformed as well. Consider the Fourier Transform $\mathcal{F}$ given in Eq. (1.11), applied to the displayed part of Eq. (2.8),

$$V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}+\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\langle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle-\langle\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\langle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle\right]$$
$$\overset{\mathcal{F}}{=}\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\sum_{\sigma}\left[\cos(\delta k_x)+\cos(\delta k_y)\right]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}-E^{(\mathrm{N})}_{\mathrm{F}/V} \tag{2.9}$$

Here, the 2 prefactor comes from recognizing that the first two operators in square brackets in the first line generate identical contributions to the full sum; the double counting energy shift is given by

$$E^{(\mathrm{N})}_{\mathrm{F}/V}\equiv\frac{V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\sum_{\sigma}\left[\cos(\delta k_x)+\cos(\delta k_y)\right]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\rangle\langle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}\rangle \tag{2.10}$$

Up to this point, this derivation is general and will be re-used in next chapters. Here the specificity of the physical phase settles in: if we assume the normal state to be a free degenerate Fermi gas with renormalized bands, we get

$$\langle\hat{c}^\dagger_{\mathbf{k}_1\sigma_1}\hat{c}_{\mathbf{k}_2\sigma_2}\rangle=\delta_{\mathbf{k}_1=\mathbf{k}_2}\delta_{\sigma_1=\sigma_2}f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})$$

being f the Fermi-Dirac distribution,

$$f(\epsilon;\beta,\mu)\equiv\frac{1}{e^{\beta(\epsilon-\mu)}+1}$$

It follows that Eq. (2.9) becomes

$$V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}+\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\langle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle-\langle\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\langle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle\right]$$
$$\overset{\mathcal{F}}{=}\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}\left[\cos(2k_x)+\cos(2k_y)\right]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}\sigma}-E^{(\mathrm{N})}_{\mathrm{F}/V} \tag{2.11}$$

since the only contribution comes from $\mathbf{k}'=-\mathbf{k}$. Let now $\mathbf{q}\equiv\mathbf{K}+\mathbf{k}$, $\mathbf{q}'\equiv\mathbf{K}-\mathbf{k}$. Being for $\ell=x,y$

$$\begin{aligned}\delta q_\ell&\equiv q_\ell-q'_\ell\\&=(K_\ell+k_\ell)-(K_\ell-k_\ell)\\&=2k_\ell\end{aligned}$$

we reduce Eq. (2.9) to

$$V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}+\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\langle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle-\langle\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\langle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle\right]$$
$$\overset{\mathcal{F}}{=}\frac{2V}{L_xL_y}\sum_{\mathbf{q},\mathbf{q}'}\sum_{\sigma}\left[\cos(\delta q_x)+\cos(\delta q_y)\right]\langle\hat{c}^\dagger_{\mathbf{q}\sigma}\hat{c}_{\mathbf{q}\sigma}\rangle\hat{c}^\dagger_{\mathbf{q}'\sigma}\hat{c}_{\mathbf{q}'\sigma}-E^{(\mathrm{N})}_{\mathrm{F}/V} \tag{2.12}$$

Recall now the result of Eq. (1.20),

$$\cos(\delta q_x)+\cos(\delta q_y)=\frac{1}{2}\sum_{\gamma}\varphi^{(\gamma)}_{\mathbf{q}}\varphi^{(\gamma)*}_{\mathbf{q}'}\quad\text{for }\gamma\in\{s^*,p_x,p_y,d_{x^2-y^2}\}$$

Thanks to this handy property, we reduce Eq. (2.12) to

$$V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}+\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\langle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle-\langle\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\langle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle\right]$$
$$\overset{\mathcal{F}}{=}\sum_{\gamma,\sigma}\left[\frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{q}}u_{\mathbf{q}\sigma}\right]\times V\sum_{\mathbf{q}'\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{q}'}\hat{c}^\dagger_{\mathbf{q}'\sigma}\hat{c}_{\mathbf{q}'\sigma}-E^{(\mathrm{N})}_{\mathrm{F}/V} \tag{2.13}$$

Operator	Sector	RWCs	Net effect
$\hat{H}_U$		Hartree	μ shift by $-nU$
$\hat{H}_V$	o.s.	Hartree	μ shift by nzV
$\hat{H}_V$	s.s.	Hartree Fock	μ shift by nzV Bands renormalization and resymmetrization

Table 2.1 | Relevant Wick's contractions for the normal phase.

where we defined the EV:

$$u_{\mathbf{q}\sigma} \equiv \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle \tag{2.14}$$

Of course, the numerical value of $u_{\mathbf{q}\sigma}$ is given by the Fermi-Dirac distribution; however, for the sake of reproducibility of this derivation in next chapters, we omit that for a moment. Let $u^{(\gamma)}_\sigma$ be its γ-wave component,

$$u^{(\gamma)}_\sigma = \frac{1}{L_x L_y} \sum_{\mathbf{q}\in \mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle$$

Now, the bare bands $\epsilon_{\mathbf{k}}$ are s^*-wave symmetric, but in general $\tilde{\epsilon}_{\mathbf{k}}$ can exhibit a more complex symmetry. Surely, due to spatial inversion invariance, it must be

$$u_{\mathbf{q}\sigma} = u_{-\mathbf{q}\sigma}$$

thus

$$u^{(p_\ell)}_\sigma = 0 \quad \text{for} \quad \ell \in \{x, y\}$$

recalling the antisymmetric form factors of Tab 1.5. However, $u^{(s^*)}_\sigma$ and $u^{(d)}_\sigma$ are in general non-zero. Eq. (2.13) becomes

$$\begin{aligned} V \sum_{\langle ij \rangle} \sum_\sigma \Big[\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle \Big] \\ \stackrel{\mathcal{F}}{=} V \sum_{\mathbf{q}\in \mathrm{BZ}} \sum_\sigma \Big[u^{(s^*)}_\sigma \varphi^{(s^*)*}_{\mathbf{q}} + u^{(d)}_\sigma \varphi^{(d)*}_{\mathbf{q}} \Big] \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \end{aligned} \tag{2.15}$$

Let

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv \epsilon_{\mathbf{k}\sigma} + u^{(s^*)}_\sigma V (\cos q_x + \cos q_y) + u^{(d)}_\sigma V (\cos q_x - \cos q_y) \tag{2.16}$$

This equation defines the bands renormalization. The fact that both a s^* and a d-wave component appear introduces an interesting anisotropy that is discusses in next section. Tab. 2.1 lists the various RWCs and their net effect on the MFT hamiltonian.

2.3.2 Fock contraction energy

Finally, let us derive the Fock contraction energy by expanding Eq. (2.10):

$$\begin{aligned} E^{(\mathrm{N})}_{\mathrm{F}/V} &= \frac{V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_\sigma \left[\cos(\delta q_x) + \cos(\delta q_y) \right] \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}'\sigma} \rangle \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}'\sigma} \rangle \\ &= L_x L_y \times \frac{V}{2} \sum_{\gamma,\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}\in \mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} u_{\mathbf{q}\sigma} \right] \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}'\in \mathrm{BZ}} \varphi'^{(\gamma)*}_{\mathbf{q}} u_{\mathbf{q}'\sigma} \right] \\ &= L_x L_y \times \frac{V}{2} \sum_\sigma \left[\left| u^{(s^*)}_\sigma \right|^2 + \left| u^{(d)}_\sigma \right|^2 \right] \end{aligned} \tag{2.17}$$

This expression gives the Fock contraction energy coming from the $\hat{H}_V$ operator. The particularly relevant result, here, is the fact that the full spectral amplitude of $u_{\mathbf{k}\sigma}$ here is weighted and contributes to contraction energy with a V linear prefactor. This suggests that a mixed-symmetry band could be energetically preferred.

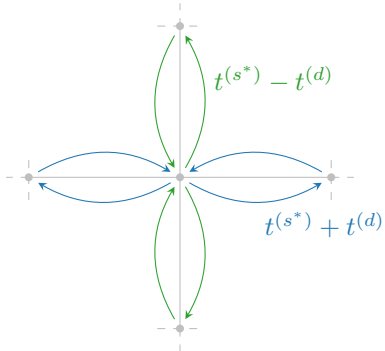


Figure 2.1 | Graphic rendition of anisotropic hopping, with a s^*-wave component (symmetric by $\pi/$ rotations) and a d-wave component (antisymmetric).

2.3.3 Extending hopping to d-wave bands

From Tabs. 1.4 and 1.5 we can inverse Fourier transform the form factors to get the full effective hopping,

$$\tilde{t}_{ij\sigma} = \sum_{\gamma\in\{s^*,s\}} t_\sigma^{(\gamma)} \varphi_{ij}^{(\gamma)}$$

with

$$t_\sigma^{(s^*)} = t - \frac{u_\sigma^{(s^*)} V}{2} \qquad t_\sigma^{(d)} = -\frac{u_\sigma^{(d)} V}{2}$$

Interestingly enough, in principle an anisotropy in the normal diffusion between x and y direction emerges. A schematics of anisotropic mixed symmetry hopping is depicted in Fig. 2.1: as is evident, for each bond linking two sites the hopping parameter can be a positive or negative combination of the two components whether the link direction is vertical or horizontal. [To be continued...]

Note that for null HFPs,

$$u_\sigma^{(s^*)} = 0 \qquad u_\sigma^{(d)} = 0$$

we have:

$$\tilde{t}_{ij\sigma} = \frac{t}{2}\left[\delta_{j=i+\delta_x} + \delta_{j=i-\delta_x} + \delta_{j=i+\delta_y} + \delta_{j=i-\delta_y}\right]$$

The fact that $t/2$ appears here instead of t is not inconsistent, it's just a consequence of the equivalence:

$$-t\sum_{\langle ij\rangle}\sum_\sigma \left[\hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + \text{h.c.}\right] = -\frac{t}{2}\sum_{i,j\in\mathcal{S}}\sum_\sigma \left[\hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + \text{h.c.}\right]$$

The diffusion RMP $\tilde{t}_{ij\sigma}$ is to be interpreted like the right-hand side of the above equation, with the kinetic operator becoming:

$$\hat{H}_{\tilde{t}} \equiv -\frac{1}{2}\sum_{i,j\in\mathcal{S}}\sum_\sigma \left[\tilde{t}_{ij\sigma}\hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + \text{h.c.}\right]$$

To limit the sum to NNs is a matter of the symmetry structure of the function $\tilde{t}_{ij\sigma}$.

2.4 Normal free energy density

An important quantity to derive in order to compare different phases is free energy density, defined as

$$f_{\text{MFT}}^{(\text{N})} \equiv \frac{F_{\text{MFT}}^{(\text{N})}}{L_x L_y}$$

being F_{MFT} the total free energy. Consider the complete hamiltonian

$$\hat{H} \simeq \sum_{\mathbf{k},\sigma} \tilde{\epsilon}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} + E_{\text{c}}^{(\text{N})} - \tilde{\mu}\hat{N}$$

where $E_{\mathrm{c}}^{(\mathrm{N})}$ is the (negative) total contraction energy for the normal phase,

$$E_{\mathrm{c}}^{(\mathrm{N})} \equiv -\left[E_{\mathrm{H}/U}^{(\mathrm{N})} + E_{\mathrm{H}/V}^{(\mathrm{N})} + E_{\mathrm{F}/V}^{(\mathrm{N})}\right]$$

having included the various contraction energies from Eqns. (2.6), (2.7) and (2.17).

Let us now take together all terms. From now on, we will omit spin index σ, making use of the always present $\mathbb{Z}_2$ spin inversion symmetry discussed in Sec. 2.1. For a free Fermi gas, the free energy density is simply obtained by summing four contributions:

1. The free particles thermal ground state energy density,

$$e_{\mathrm{g}}^{(\mathrm{N})} = \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

2. The contraction energy density,

$$\begin{aligned} e_{\mathrm{c}}^{(\mathrm{N})} &\equiv \frac{E_{\mathrm{c}}^{(\mathrm{N})}}{L_x L_y} \\ &= -n^2 U + V\left[2zn^2 - \left|u^{(s^*)}\right|^2 - \left|u^{(d)}\right|^2\right] \end{aligned}$$

3. The entropic free energy

$$e_s^{(\mathrm{N})} \equiv -Ts$$

with $T = 1/k_B\beta$ and s the free Fermi gas entropy density for a single spin sector [**grosso2014solid**],

$$s = -\frac{k_B}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \left[(1 - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})) \ln(1 - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})) + f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \ln f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})\right] \tag{2.18}$$

We use the identity:

$$\ln\left(1 - \frac{1}{e^x + 1}\right) = x + \ln\left(\frac{1}{e^x + 1}\right)$$

and reduce the above expression to:

$$\begin{aligned} s = -\frac{k_B}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \Big[&\ln(1 - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})) + f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \ln f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \\ &- \beta(\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \ln f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})\Big] \end{aligned}$$

Second and fourth terms cancel out, and multiplying by $-2T$ (taking care of spin degeneracy) we have

$$e_s^{(\mathrm{N})} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \ln(1 - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})) - \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

4. The chemical potential correction

$$e_n^{(\mathrm{N})} = \mu n$$

Note here that we are using the "physical" value of μ, not its RMP counterpart $\tilde{\mu}$. The reason for this is discussed in Sec. 1.4.4.

Thus, composing everything

$$\begin{aligned} f_{\mathrm{MFT}}^{(\mathrm{N})} &\equiv e_{\mathrm{g}}^{(\mathrm{N})} + e_{\mathrm{c}}^{(\mathrm{N})} + e_s^{(\mathrm{N})} + e_n^{(\mathrm{N})} \\ &= \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \ln(1 - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})) - n^2 U + V\left[2zn^2 - \left|u^{(s^*)}\right|^2 - \left|u^{(d)}\right|^2\right] + \mu n \end{aligned} \tag{2.19}$$

When it comes to algorithmic computation of this quantity, it is important to express μ in terms of its RMP counterpart $\tilde{\mu}$, as discussed in Sec. 1.4.4. By doing this we are considering all the effects of MFT decomposition of $\hat{H}$.

2.5 Numerical simulations

[To be continued...]

2.5.1 HF iterative algorithm sketch

We are finally able to move to the computational part of this chapter. The aim is to write down an algorithm of the kind of Sec. 1.4.1. For the normal phase, the HFPs vector has two components:

$$\mathbf{v} = \begin{bmatrix} u^{(s^*)} \\ u^{(d)} \end{bmatrix}$$

As before, we treat spin as a pure degeneracy. The HFPs are characterized by the self-consistency equations

$$u^{(s^*)} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \tag{2.20}$$

$$u^{(d)} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{BZ}} (\cos k_x - \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \tag{2.21}$$

The resulting model has two RMPs:

$$\tilde{t}_{ij} \equiv \left(t - \frac{u^{(s^*)} V}{2}\right) \varphi_{ij}^{(s^*)} - \frac{u^{(d)} V}{2} \varphi_{ij}^{(d)} \tag{2.22}$$

$$\tilde{\mu} \equiv \mu + n(2zV - U) \tag{2.23}$$

the second one being only dependent on the model parameters and not on HFPs. We can define the direction-dependent hopping as

$$\tilde{t}^{(x)} \equiv t - \frac{u^{(s^*)} - u^{(d)}}{2} V \tag{2.24}$$

$$\tilde{t}^{(y)} \equiv t - \frac{u^{(s^*)} + u^{(d)}}{2} V \tag{2.25}$$

Both components are controlled by V. Evidently $u^{(s^*)}$ is a measure of the direction-averaged hopping shift with respect to V, while $u^{(d)}$ measures the hopping anisotropy with respect to V. Finally, the phase free energy is given by Eq. (2.19). [To be continued...]

Chapter 3

Anti-Ferromagnetic instability

This chapter is devoted to an in-depth discussion of the AF long-range ordering in the EHM by the means of MFT. In the context of the 2D HM, discussed in App. **??**, AF ordering establishes as an unconventional metallic phase which is only stable at half-filling (as is discussed in Sec. **??**). The situation for the EHM is similar: the present analysis is carried out based on the argument that, given the symmetry structure of the non-local NN interaction, the fundamental features of the commensurate AF phase of the HM are effectively preserved, while the parameters are redefined by the interaction itself. The two most relevant effects we discuss in this chapter are the "gap complexification" (Sec. [Missing]) and the "hopping renormalization" (Sec. [Missing]), the latter being a general feature of the EHM common to all phases preserving (or, at most, weakening without destroying) the translational invariance, discussed in Sec. 2.3.

3.1 The AF phase and its symmetries

When discussing the AF phase on the EHM square lattice, we must specify the type of antiferromagnetism we are dealing with. SDW antiferromagnets on a square lattice can emerge in a variety of ways, the simplest one – described in this text – being the so-called "commensurate". By this we mean essentially a kind of long range spin density ordering varying on the length scale of two lattice sites. This type of AF structure is particularly interesting on a square lattice because the phase transition is simply described in reciprocal space by the order parameter

$$\hat{\psi}_{\mathbf{k}\sigma} \equiv \hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma}$$

going from a null EV in the normal phase to a non-zero value. This phenomenon occurs at the so called "AF vector"

$$\boldsymbol{\pi} \equiv (\pi, \pi)$$

Large portions of a square lattice Fermi surface (FS) can turn out to be connected precisely by an AF vector: this implies that, if the hamiltonian makes available a scattering channel at this wavevector, electrons can jump from one side to the other, making said EV non-zero.

In principle, the AF phase is physically established by breaking the SU(2) spin rotation symmetry down to the $\mathrm{U}^z(1)$ projected spin rotation symmetry; however, it is to be noted that – even if the concept of SSB is well defined only for the $\mathrm{SU}(2) \to \mathrm{U}^z(1)$ symmetry reduction – the hamiltonian in thermodynamic limit remains $\mathbb{Z}_2$ invariant with respect to spin inversion, $\sigma \leftrightarrow \overline{\sigma}$. Identical considerations as in Sec. 2.1 hold This: all physical quantities are spin-independent. As for the normal phase, we will use this considerations at the end of the calculations.

3.1.1 Antiferromagnetism in the HM

In this short section we recall the principal feature of the AF phase established in the conventional HM, extensively discussed in Sec. **??**. Before starting, let us define as in Eqns. (**??**) and (**??**) the AF and density fluctuations operators,

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} \tag{3.1}$$

$$\hat{\psi}_{\mathbf{k}\sigma} \equiv \hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \tag{3.2}$$

EV	Meaning	Equation
$\langle \hat{s}^x_{\mathbf{k}\sigma} \rangle$	$\left\langle \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}^\dagger_{\mathbf{k}\sigma} \right\rangle$	(??)
$i \langle \hat{s}^y_{\mathbf{k}\sigma} \rangle$	$\left\langle \hat{\psi}_{\mathbf{k}\sigma} - \hat{\psi}^\dagger_{\mathbf{k}\sigma} \right\rangle$	(??)
$\langle \hat{s}^z_{\mathbf{k}\sigma} \rangle$	$\langle \hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle$	(??)

Table 3.1 | List of the various expectation values (EVs) of each pseudospin component, their meaning in terms of fermionic operators and the reference equations from the pure HM.

which represent the two essential fermionic quadratic operators of the analytic MFT derivation. Within the pure HM, the AF phase is specified by the Ansatz (**??**) which is explicitly breaking translational invariance in each spin sector, while preserving $\mathrm{U}^z(1)$ and $\mathrm{U}^c(1)$ symmetries. As reported in Tab. 1.2, the only RWC comes in Hartree channel, since the local two-body interaction couples operators at opposite spin index. The hamiltonian $\hat{H}$ is reduced thereby to the form of Eq. (**??**)

$$\hat{H}_t + \hat{H}_U \overset{\text{MFT}}{\simeq} -t \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_\sigma \left[\hat{c}^\dagger_{\mathbf{r}\sigma} \hat{c}_{\mathbf{r}'\sigma} + \text{h.c.} \right] + nU \sum_{\mathbf{r}} \left[\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow} \right] - mU \sum_{\mathbf{r}} (-1)^{x+y} \left[\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow} \right] - E^{(\text{AF})}_{\text{H}/U}$$

with $E^{(\text{AF})}_{\text{H}/U}$ the contraction energy resulting from double counting terms, given explicitly in Eq. (**??**). In reciprocal space, the hamiltonian decomposes as in Eq. (**??**),

$$\hat{H}_t + \hat{H}_U \overset{\text{MFT}}{\simeq} \sum_{\mathbf{k}\in\text{MBZ}} \sum_\sigma \hat{\Psi}^\dagger_{\mathbf{k}\sigma} h_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} - E^{(\text{AF})}_{\text{H}/U} \quad \text{being} \quad h_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \epsilon_{\mathbf{k}\sigma} & -\Delta_{\mathbf{k}\sigma} \\ -\Delta^*_{\mathbf{k}\sigma} & -\epsilon_{\mathbf{k}\sigma} \end{bmatrix}$$

and $\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} mU$. Nambu spinorial formulation is used,

$$\hat{\Psi}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\sigma} \\ \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \end{bmatrix}$$

and the free electrons energy is simply the tight binding energy

$$\epsilon_{\mathbf{k}\sigma} = -2t \left(\cos k_x + \cos k_y \right)$$

which is spin-invariant. The MFT description of the model reduces to a gas of free "γ-fermions", described by the Nambu spinor of Eq. (**??**),

$$\hat{\Gamma}_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} = \begin{bmatrix} \hat{\gamma}^{(+)}_{\mathbf{k}\sigma} \\ \hat{\gamma}^{(-)}_{\mathbf{k}\sigma} \end{bmatrix}$$

where

$$W_{\mathbf{k}\sigma} = \begin{bmatrix} \cos\theta_{\mathbf{k}\sigma} & -\sin\theta_{\mathbf{k}\sigma} \\ \sin\theta_{\mathbf{k}\sigma} & \cos\theta_{\mathbf{k}\sigma} \end{bmatrix} \quad \text{and} \quad \sin 2\theta_{\mathbf{k}\sigma} \equiv \frac{\Delta_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}}$$

These fermions populate the two bands $\pm E_{\mathbf{k}\sigma} = \sqrt{\epsilon^2_{\mathbf{k}\sigma} + |\Delta_{\mathbf{k}\sigma}|^2}$. The entire system is mapped onto an ensemble of pseudo-spins, each subject to a pseudo-field, as in Fig. 3.1. To diagonalize the system essentially means to anti-align each pseudo-spin with the z axis. Each pseudospin component expectation value (EV) is described by a simple equation and has a precise meaning, listed in Tab. 3.1.

3.1.2 Extension to the EHM

Consider now the non-local interaction $\hat{H}_V$: since only translational invariance is partially broken in the AF phase, the only RWCs within the MFT decomposition are Hartree terms and the same-spin Fock term. The discussion is thus essentially the same as for the normal phase of Chap. 2, with

the only relevant difference being the Ansatz we use in Eq. (2.1), substituted by the commensurate oscillatory Ansatz of Eq. (**??**). As for the normal phase, by design the MFT solution to the model for the given phase is described by a redefinition of the various physical quantities and mathematical objects,

$$\epsilon_{\mathbf{k}\sigma} \to \tilde{\epsilon}_{\mathbf{k}\sigma} \qquad E_{\mathbf{k}\sigma} \to \tilde{E}_{\mathbf{k}\sigma} \qquad \Delta_{\mathbf{k}\sigma} \to \tilde{\Delta}_{\mathbf{k}\sigma} \qquad \mu \to \tilde{\mu}$$

The band energies renormalization is simply

$$\tilde{E}_{\mathbf{k}\sigma} \equiv \sqrt{\tilde{\epsilon}^2_{\mathbf{k}\sigma} + |\tilde{\Delta}_{\mathbf{k}\sigma}|^2}$$

and the following relations hold (the *tilde* sign indicates the renormalized quantity):

$$\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \tau^x \hat{\Psi}_{\mathbf{k}\sigma} \rangle = -\sin(2\tilde{\theta}_{\mathbf{k}\sigma}) \sin(2\tilde{\zeta}_{\mathbf{k}\sigma}) \langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \tag{3.3}$$

$$\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \tau^y \hat{\Psi}_{\mathbf{k}\sigma} \rangle = -\sin(2\tilde{\theta}_{\mathbf{k}\sigma}) \cos(2\tilde{\zeta}_{\mathbf{k}\sigma}) \langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \tag{3.4}$$

$$\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Psi}_{\mathbf{k}\sigma} \rangle = \cos(2\tilde{\theta}_{\mathbf{k}\sigma}) \langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \tag{3.5}$$

equivalent to the ones for the pure HM, Eqns. (**??**), (**??**) and (**??**). Moving to the the tilted frame, the z component of the pseudospin is just a thermal superposition of the "up" and "down" states

$$\langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle = \frac{1}{2} \left[f\left(\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) - f\left(-\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) \right] \tag{3.6}$$

Let us define the (renormalized) thermal factors as in Eq. (**??**),

$$\tilde{\mathrm{Th}}^{(\pm)}_{\mathbf{k}\sigma}[\beta, \tilde{\mu}] \equiv f\left(\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) \pm f\left(-\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) \tag{3.7}$$

The angles satisfy:

$$\sin(2\tilde{\xi}_{\mathbf{k}\sigma}) = \frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{|\tilde{\Delta}_{\mathbf{k}\sigma}|} \quad \cos(2\tilde{\xi}_{\mathbf{k}\sigma}) = \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{|\tilde{\Delta}_{\mathbf{k}\sigma}|} \quad \sin(2\tilde{\theta}_{\mathbf{k}\sigma}) = \frac{|\tilde{\Delta}_{\mathbf{k}\sigma}|}{\tilde{E}_{\mathbf{k}\sigma}} \quad \cos(2\tilde{\theta}_{\mathbf{k}\sigma}) = \frac{\tilde{\epsilon}_{\mathbf{k}\sigma}}{\tilde{E}_{\mathbf{k}\sigma}} \tag{3.8}$$

The physical behaviour is the same as for the pure HM. In terms of MFT discussion, the AF phase is similar enough to the normal phase of Chap. 2: the RWCs are the same of Sec. **??**, listed in Tabs. 1.2 and 1.3. All three Hartree contractions, responsible in the normal phase of the chemical potential shift, and the s.s. Fock contraction, that originated the hopping renormalization. Let us start by the Hartree terms: due to the "staggered" nature of the commensurate AF phase, their role is here more complex. Then we will move to the Fock term, discussing the MFT approximations effect on bands. Next sections are basically an extension of the calculations of Chap. 2.

3.2 Non-local Hartree effects: chemical potential renormalization

As in Eq. (2.3), the same-spin and opposite-spin non-local Hartree terms are

$$\hat{H}_V \simeq \overbrace{-V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{n}_{i\sigma} \rangle \hat{n}_{j\sigma} + \hat{n}_{i\sigma} \langle \hat{n}_{j\sigma} \rangle - \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\sigma} \rangle \right]}^{\text{s.s.}}$$
$$\underbrace{-V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{n}_{i\sigma} \rangle \hat{n}_{j\bar{\sigma}} + \hat{n}_{i\sigma} \langle \hat{n}_{j\bar{\sigma}} \rangle - \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\bar{\sigma}} \rangle \right]}_{\text{o.s.}} + (\text{all the rest})$$

Let $i \to \mathbf{r} = (x, y)$ and $j \to \mathbf{r}' = (x', y')$. The key difference in the present derivation, with respect to the normal phase, is the single-site density Ansatz: instead of Eq. (2.1), we use Eq. (**??**), which describes an oscillatory density for each spin sector,

$$\langle \hat{n}_{\mathbf{r}\sigma} \rangle = n - (-1)^{x+y+\delta_{\sigma=\uparrow}} m$$

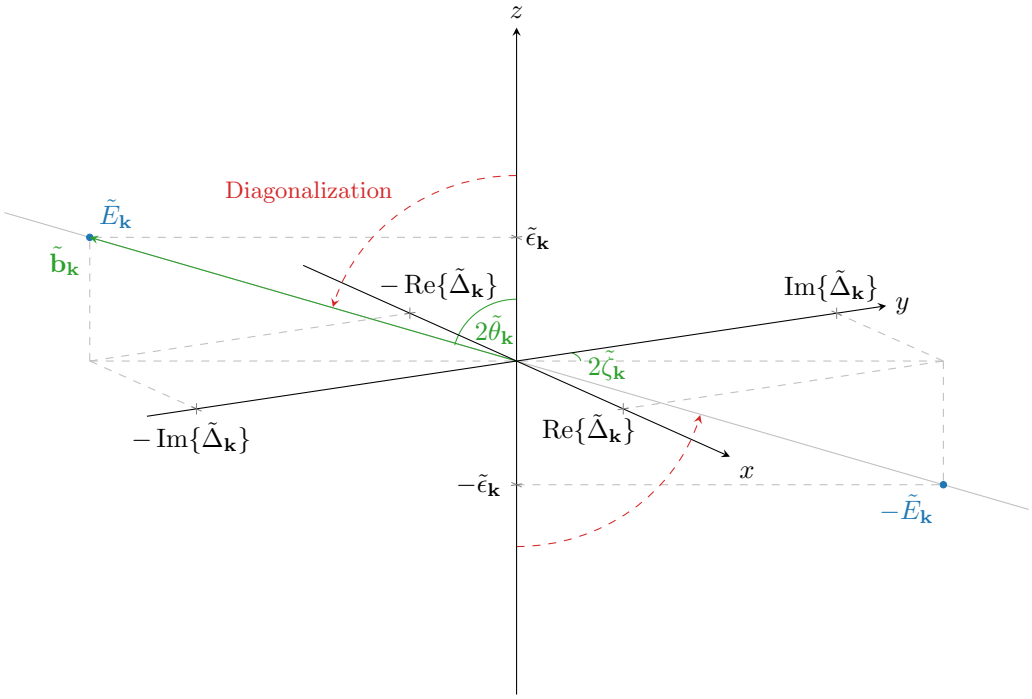


Figure 3.1 Sketch of the diagonalization of the pseudo-spin problem. The dashed line represents the diagonalization given by the z-axis alignment with the field direction. In the top-right side of the picture are listed the expectation values of the original Nambu spinor. For simplicity everywhere the index $\sigma = \uparrow$ has been omitted.

This Ansatz is composed of two parts: the offset n, and the oscillatory part m. The algebraic derivation relative to the n part is, of course, the same as for the normal phase. We now treat separately the "operatorial" part of the braced terms of Eq. (2.3),

$$-V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\langle \hat{n}_{\mathbf{r}\sigma} \rangle \hat{n}_{\mathbf{r}'\sigma} + \hat{n}_{\mathbf{r}\sigma} \langle \hat{n}_{\mathbf{r}'\sigma} \rangle] - V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\langle \hat{n}_{\mathbf{r}\sigma} \rangle \hat{n}_{\mathbf{r}'\overline{\sigma}} + \hat{n}_{\mathbf{r}\sigma} \langle \hat{n}_{\mathbf{r}'\overline{\sigma}} \rangle] \tag{3.9}$$

as well as the its scalar counterpart

$$V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \langle \hat{n}_{\mathbf{r}'\sigma} \rangle + V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \langle \hat{n}_{\mathbf{r}'\overline{\sigma}} \rangle \tag{3.10}$$

which accounts for the energy shift to be accounted when performing Wick's contractions.

3.2.1 Hartree channel in the AF phase: analytical derivation

Let us start from the operator of Expr. (3.9). By inserting the Ansatz of Eq. (**??**), we get

$$\overbrace{-nV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{n}_{\mathbf{r}'\sigma} + \hat{n}_{\mathbf{r}\sigma}] + mV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} (-1)^{\delta_{\sigma=\uparrow}} \left[(-1)^{x+y} \hat{n}_{\mathbf{r}'\sigma} + (-1)^{x'+y'} \hat{n}_{\mathbf{r}\sigma} \right]}^{\text{s.s.}}$$
$$\underbrace{-nV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{n}_{\mathbf{r}'\overline{\sigma}} + \hat{n}_{\mathbf{r}\sigma}] + mV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \left[(-1)^{x+y+\delta_{\sigma=\uparrow}} \hat{n}_{\mathbf{r}'\overline{\sigma}} + (-1)^{x'+y'+\delta_{\overline{\sigma}=\uparrow}} \hat{n}_{\mathbf{r}\sigma} \right]}_{\text{o.s.}}$$

For a square lattice, if $\mathbf{r} = (x, y)$ and $\mathbf{r}' = (x', y')$ are NNs, evidently

$$(-1)^{x'+y'} = (-1)^{x+y+1}$$

Moreover,

$$(-1)^{\delta_{\overline{\sigma}=\uparrow}} = (-1)^{\delta_{\sigma=\uparrow}+1}$$

We obtain

$$\overbrace{-nV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}[\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\sigma}]}^{\text{s.s. (density)}}-\overbrace{mV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(-1)^{x+y+\delta_{\sigma=\uparrow}}[\hat{n}_{\mathbf{r}\sigma}-\hat{n}_{\mathbf{r}'\sigma}]}^{\text{s.s. (magnetization)}}$$
$$\underbrace{-nV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}[\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\overline{\sigma}}]}_{\text{o.s. (density)}}+\underbrace{mV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(-1)^{x+y+\delta_{\sigma=\uparrow}}[\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\overline{\sigma}}]}_{\text{o.s. (magnetization)}} \tag{3.11}$$

In the above expressions the various contribution have been separated in "density" contributions and "magnetization" contributions. Let us deal with these separately.

Density terms. Consider the s.s. and o.s. (density) terms of Expr. (3.11),

$$-nV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}[\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\sigma}]-nV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}[\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\overline{\sigma}}] \tag{3.12}$$

The discussion is identical to the one of Sec. 2.2.1 for the normal phase, with the renormalization of μ being the same of Eq. (2.5).

$$\tilde{\mu} \equiv \mu + n(2zV - U) \tag{3.13}$$

Within this equation we are also already implicitly considering the μ shift effect due to $\hat{H}_U$.

Magnetization terms. The s.s. and o.s. (magnetization) terms of Expr. (3.11) perfectly cancel out:

$$mV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(-1)^{x+y+\delta_{\sigma=\uparrow}}[(\hat{n}_{\mathbf{r}\sigma}-\hat{n}_{\mathbf{r}'\sigma})-(\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\overline{\sigma}})]=0$$

thus the only net effect of the Hartree RWCs affects μ. This is somewhat to be expected: in the context of the HM, the gap function emerges in its purely s-wave structure from this precise computation on Hartree terms. However, the $\hat{H}_V$ non-local interaction cannot provide a pure s-wave potential component. Effects on the gap function are to be observed somewhere else.

Let us now move to the last part of the Hartree effects on the AF phase originated by $\hat{H}_V$, which is, the energy shift due to double counting terms originated by MFT decomposition of the quartic interactions $\hat{H}_U$ and $\hat{H}_V$.

3.2.2 Hartree contraction energy

Consider the scalar Expr. (3.10): it's the sum of two parts, the first coming from s.s. sector, the second from o.s. sector.

s.s. sector double counting energy. For the s.s. part, explicitating the two spin sectors, we get

$$V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\langle\hat{n}_{\mathbf{r}\sigma}\rangle\langle\hat{n}_{\mathbf{r}'\sigma}\rangle = \overbrace{V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\Big[(n+(-1)^{x+y}m)(n+(-1)^{x'+y'}m)\Big]}^{\sigma=\uparrow}$$
$$\underbrace{+V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\Big[(n-(-1)^{x+y}m)(n-(-1)^{x'+y'}m)\Big]}_{\sigma=\downarrow}$$

Evidently the mixed terms (those of order $\mathcal{O}(n)\times\mathcal{O}(m)$) cancel out, and considering that the sign factor is necessarily alternating on neighboiring site, the remainder is just

$$V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\sigma}\rangle = V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}2(n^2-m^2)$$

o.s. sector double counting energy. Proceeding identically on the o.s. sector, we get

$$V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\langle \hat{n}_{\mathbf{r}\sigma}\rangle\langle \hat{n}_{\mathbf{r}'\overline{\sigma}}\rangle = \overbrace{V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\Big[(n+(-1)^{x+y}m)(n-(-1)^{x'+y'}m)\Big]}^{\sigma=\uparrow} + \underbrace{V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\Big[(n-(-1)^{x+y}m)(n+(-1)^{x'+y'}m)\Big]}_{\sigma=\downarrow}$$

which gives immediately

$$V\sum_{\langle ij\rangle}\sum_{\sigma}\langle \hat{n}_{i\sigma}\rangle\langle \hat{n}_{j\overline{\sigma}}\rangle = V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}2(n^2+m^2)$$

Hence, the total "contraction energy", given by the sum of both contributions, is identical to the normal one of Eq. (2.7) indeed:

$$E^{(\mathrm{AF})}_{\mathrm{H}/V} = -2V\sum_{\langle ij\rangle}\sum_{\sigma}n^2 = 4n^2V\times\frac{z}{2}L_xL_y \tag{3.14}$$

the minus sign coming from the fact that we are subtracting this energy shift. This implies that, when comparing normal and AF phase, we can rule out this energy contribution, being equal for both free energy densities.

3.3 Fock effects: gap and bands renormalization

We explored thoroughly the Hartree contractions effects on the MFT hamiltonian. Let us now move to the effects originated by Fock contractions. We can use the derivation of Sec. 2.3 up to Eq. (2.11),

$$\begin{aligned} V\sum_{\langle ij\rangle}\sum_{\sigma}&\Big[\langle \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}+\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\langle \hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle-\langle \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\langle \hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle\Big] \\ &\overset{\mathcal{F}}{=}\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}\left[\cos(2k_x)+\cos(2k_y)\right]\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}\sigma}-E^{(\mathrm{AF})}_{\mathrm{F}/V} \end{aligned} \tag{3.15}$$

where $E^{(\mathrm{AF})}_{\mathrm{F}/V}$ is defined as in Eq. (2.10), with the expectation value being computed on the AF (thermal) ground-state. As before, we discuss the operatorial part and the scalar part separately.

3.3.1 Fock channel in the AF phase: analytical derivation

The commensurate AF phase is essentially described by two EVs being in principle non-zero,

$$\left|\langle \hat{n}_{\mathbf{k}\sigma}\rangle\right|\geq 0 \qquad \left|\langle \hat{\psi}_{\mathbf{k}\sigma}\rangle\right|\geq 0$$

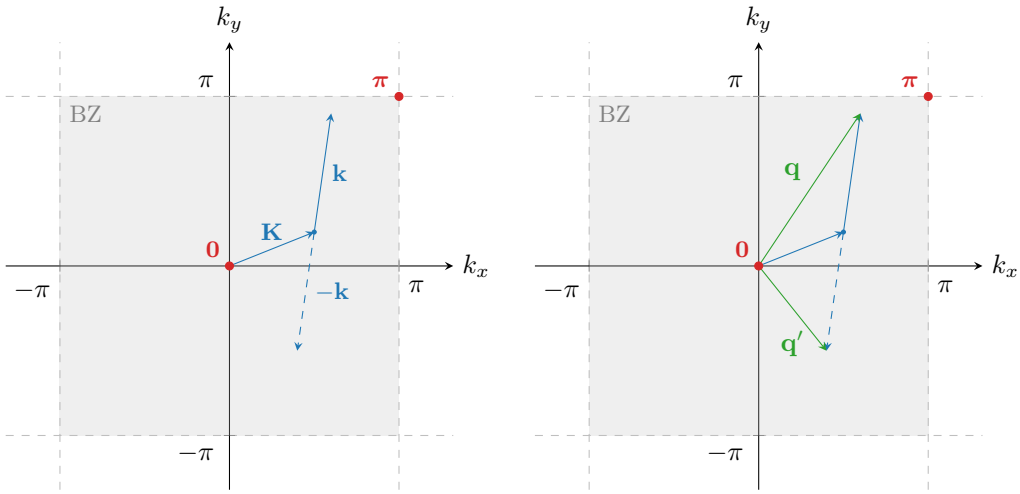
The AF ground state is a degenerate Fermi gas of $\gamma^{(\pm)}$ quasiparticles, and for null temperature acquires the simple form of Eq. (**??**). Going back to Eq. (3.15), this implies only two contributions are non-zero:

$$\mathbf{k}=-\mathbf{k}' \quad \text{or} \quad \mathbf{k}+\boldsymbol{\pi}=-\mathbf{k}'$$

The first case gives us the normal fermionic EV, $\langle \hat{n}_{\mathbf{k}\sigma}\rangle$, the second the staggered AF order parameter, $\langle \hat{\psi}_{\mathbf{k}\sigma}\rangle$. Then Eq. (3.15) is reduced to:

$$\begin{aligned}(\cdots)\overset{\mathcal{F}}{=}\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}&\left[\cos(2k_x)+\cos(2k_y)\right] \\ &\Big[\underbrace{\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}\sigma}}_{\text{Diagonal terms}}-\underbrace{\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}}_{\text{Off-diagonal terms}}\Big]\end{aligned} \tag{3.16}$$

Now, the above equation presents *diagonal* and *off-diagonal* terms.



a | Generic vectors. b | Change of variables.

Figure 3.2 | Representation of the vectors involved in the diagonal terms of Eq. (**??**). In Fig. 3.2a generic vectors are considered, cycling over all values of $\mathbf{K}, \mathbf{k} \in$ BZ. In Fig. 3.2b is depicted the variables change to the new vectors $\mathbf{q}, \mathbf{q}'$.

Diagonal terms. The diagonal terms of Eq. (3.16) are simple density interactions with the mean density field. Consider Fig. 3.2a: density at vector $\mathbf{q} \equiv \mathbf{K} + \mathbf{k}$ interacts with the mean density at vector $\mathbf{q}' \equiv \mathbf{K} - \mathbf{k}$. These variables are depicted in Fig. 3.2b. Apply in the diagonal part of Eq. (3.16) this variables change (which is the same of Eq. (2.12))

$$\begin{aligned}
\frac{2V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma} \\
&= \frac{2V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_{\sigma} [\cos(\delta q_x) + \cos(\delta q_y)] \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} \\
&\overset{*}{=} \sum_{\gamma,\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}} \varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{n}_{\mathbf{q}\sigma} \rangle \right] \times V \sum_{\mathbf{q}'} \varphi^{(\gamma)*}_{\mathbf{q}'} \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma}
\end{aligned} \tag{3.17}$$

where the sign $\overset{*}{=}$ indicates that we have substituted the decomposition of Eq. (1.20). Define as in Eq. (2.14) the EV

$$u_{\mathbf{q}\sigma} \equiv \langle \hat{n}_{\mathbf{q}\sigma} \rangle \tag{3.18}$$

The expression in square brackets of Eq. (3.17) is just its γ component, and is expanded in

$$\begin{aligned}
u^{(\gamma)}_{\sigma} &= \frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{n}_{\mathbf{q}\sigma} \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \left[\varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{n}_{\mathbf{q}\sigma} \rangle + \varphi^{(\gamma)}_{\mathbf{q}+\boldsymbol{\pi}} \langle \hat{n}_{\mathbf{q}+\boldsymbol{\pi}\sigma} \rangle \right] \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \varphi^{(\gamma)}_{\mathbf{q}} \left[\langle \hat{n}_{\mathbf{q}\sigma} \rangle - \langle \hat{n}_{\mathbf{q}+\boldsymbol{\pi}\sigma} \rangle \right] \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{\Psi}^\dagger_{\mathbf{q}\sigma} \tau^z \hat{\Psi}_{\mathbf{q}\sigma} \rangle \\
&= \frac{1}{2L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \varphi^{(\gamma)}_{\mathbf{q}} \frac{\tilde{\epsilon}_{\mathbf{q}\sigma}}{\tilde{E}_{\mathbf{q}\sigma}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta, \tilde{\mu}]
\end{aligned} \tag{3.19}$$

where, in the last passage, Eqns. (3.5), (3.6) and (3.7) have been used. Now, the AF phase does not break inversion symmetry: this implies that, whatever the shape of the renormalized bands, it

must be

$$\tilde{\epsilon}_{\mathbf{q}\sigma} = \tilde{\epsilon}_{-\mathbf{q}\sigma} \quad \Longrightarrow \quad u_\sigma^{(p_\ell)} = 0 \quad \text{with} \quad \ell \in \{x, y\}$$

The renormalized bands need to be symmetric functions of the moment. Hence for $\gamma \in \{p_x, p_y\}$ the above sum vanishes. Thus, Eq. (**??**) becomes

$$\frac{2V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_\sigma \left[\cos(2k_x) + \cos(2k_y)\right] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}$$
$$= V \sum_{\mathbf{q},\sigma} \left[u_\sigma^{(s^*)} (\cos q_x + \cos q_y) + u_\sigma^{(d)} (\cos q_x - \cos q_y) \right] \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma}$$

These two factors, together, define the full bands renormalization

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv \epsilon_{\mathbf{k}\sigma} + u_\sigma^{(s^*)} V (\cos q_x + \cos q_y) + u_\sigma^{(d)} V (\cos q_x - \cos q_y) \tag{3.20}$$

as was for the normal phase in Eq. (2.16). In Sec. 2.3.3 we discussed the appearance of anisotropic hopping with d-wave extension of the bands: here the theory predicts something identical. These RMPs are, as before, controlled linearly by V and in principle can lead to relevant changes in the kinetic sector of the hamiltonian.

Through this decomposition we have identified two HFPs and their self-consistency equations,

$$u_\sigma^{(s^*)} = \frac{1}{2L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} (\cos q_x + \cos q_y) \frac{\tilde{\epsilon}_{\mathbf{q}\sigma}}{\tilde{E}_{\mathbf{q}\sigma}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta, \tilde{\mu}] \tag{3.21}$$

$$u_\sigma^{(d)} = \frac{1}{2L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} (\cos q_x - \cos q_y) \frac{\tilde{\epsilon}_{\mathbf{q}\sigma}}{\tilde{E}_{\mathbf{q}\sigma}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta, \tilde{\mu}] \tag{3.22}$$

The general HF scheme, described in Sec. 1.4.1, is iterative and aims to use these equations iteratively to find a stable solution for these parameters.

Off-diagonal terms. Consider the off-diagonal terms of Eq. (3.16). These contribute instead to the full gap, being out of diagonal in the 2×2 hamiltonian matrix. In terms of the pseudospin picture, they account for a transverse field. Define $\mathbf{q}$, $\mathbf{q}'$ as in Fig. 3.2b, and rewrite as for Eq. (3.17)

$$-\frac{2V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_\sigma [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}$$
$$= -\sum_{\gamma,\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}} \varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \right] \times V \sum_{\mathbf{q}'} \varphi^{(\gamma)*}_{\mathbf{q}'} \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'+\boldsymbol{\pi}\sigma} \tag{3.23}$$

Proceed as done previously: define the expectation value

$$v_{\mathbf{q}\sigma} \equiv i \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \tag{3.24}$$

(which is just the AF order parameter with a complex factor) and its γ-wave component, with $\gamma \in \{s^*, d, p_x, p_y\}$:

$$\begin{aligned}
v_\sigma^{(\gamma)} &= \frac{i}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \\
&= \frac{i}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \left[\varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle + \varphi^{(\gamma)}_{\mathbf{q}+\boldsymbol{\pi}} \langle \hat{\psi}_{\mathbf{q}+\boldsymbol{\pi}\sigma} \rangle \right] \\
&= \frac{i}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \varphi^{(\gamma)}_{\mathbf{q}} \left[\langle \hat{\psi}_{\mathbf{q}\sigma} \rangle - \langle \hat{\psi}^\dagger_{\mathbf{q}\sigma} \rangle \right] \\
&= -\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{\Psi}^\dagger_{\mathbf{q}\sigma} \tau^y \hat{\Psi}_{\mathbf{q}\sigma} \rangle \\
&= \frac{1}{2L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \varphi^{(\gamma)}_{\mathbf{q}} \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{q}\sigma}\}}{\tilde{E}_{\mathbf{q}\sigma}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta, \tilde{\mu}]
\end{aligned} \tag{3.25}$$

where we made use of the relations of Tab. 3.1 as well as Eq. (3.4). We also used the obvious facts that $\langle\hat{\psi}_{\mathbf{q}+\boldsymbol{\pi}\sigma}\rangle = \langle\hat{\psi}^\dagger_{\mathbf{k}\sigma}\rangle$ and $\varphi^{(\gamma)}_{\mathbf{q}+\boldsymbol{\pi}} = -\varphi^{(\gamma)}_{\mathbf{q}}$. From the definitions of the structure factors of Tab. 1.5, it's immediate to see that:

$$v^{(s^*)}_\sigma, v^{(d)}_\sigma \in \mathbb{R} \qquad v^{(p_x)}_\sigma, v^{(p_y)}_\sigma \in i\mathbb{R}$$

Getting back to Eq. (3.23) and inserting these components, we get

$$-\frac{2V}{L_xL_y}\sum_{\mathbf{q},\mathbf{q}'}\sum_\sigma[\cos(2k_x)+\cos(2k_y)]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma} = iV\sum_{\gamma,\sigma}v^{(\gamma)}_\sigma\sum_{\mathbf{q}\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{q}}\hat{\psi}_{\mathbf{q}\sigma} \tag{3.26}$$

We can expand on the MBZ as

$$-\frac{2V}{L_xL_y}\sum_{\mathbf{q},\mathbf{q}'}\sum_\sigma[\cos(2k_x)+\cos(2k_y)]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma} = iV\sum_{\gamma,\sigma}\sum_{\mathbf{q}\in\mathrm{MBZ}}\left[v^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}}\hat{\psi}_{\mathbf{q}\sigma}+v^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}+\boldsymbol{\pi}}\hat{\psi}^\dagger_{\mathbf{q}\sigma}\right] \tag{3.27}$$

Recalling Tab. 1.5, for the symmetries we are considering

$$iv^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}} \in i\mathbb{R}$$

Moreover, since $\varphi^{(\gamma)}_{\mathbf{q}} = -\varphi^{(\gamma)}_{\mathbf{q}+\boldsymbol{\pi}}$, we have

$$iv^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}+\boldsymbol{\pi}} = \left(iv^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}}\right)^*$$

Therefore

$$-\frac{2V}{L_xL_y}\sum_{\mathbf{q},\mathbf{q}'}\sum_\sigma[\cos(2k_x)+\cos(2k_y)]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma} = V\sum_{\gamma,\sigma}\sum_{\mathbf{q}\in\mathrm{MBZ}}\left[iv^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}}\hat{\psi}_{\mathbf{q}\sigma}+\mathrm{h.c.}\right] \tag{3.28}$$

Four more real HFPs are then identified, with their self-consistency equations,

$$v^{(s^*)}_\sigma = \frac{1}{2L_xL_y}\sum_{\mathbf{q}\in\mathrm{MBZ}}(\cos q_x+\cos q_y)\frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{q}\sigma}\}}{\tilde{E}_{\mathbf{q}\sigma}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta,\tilde{\mu}] \tag{3.29}$$

$$v^{(d)}_\sigma = \frac{1}{2L_xL_y}\sum_{\mathbf{q}\in\mathrm{MBZ}}(\cos q_x-\cos q_y)\frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{q}\sigma}\}}{\tilde{E}_{\mathbf{q}\sigma}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta,\tilde{\mu}] \tag{3.30}$$

$$-iv^{(p_x)}_\sigma = \frac{1}{\sqrt{2}L_xL_y}\sum_{\mathbf{q}\in\mathrm{MBZ}}\sin q_x\frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{q}\sigma}\}}{\tilde{E}_{\mathbf{q}\sigma}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta,\tilde{\mu}] \tag{3.31}$$

$$-iv^{(p_y)}_\sigma = \frac{1}{\sqrt{2}L_xL_y}\sum_{\mathbf{q}\in\mathrm{MBZ}}\sin q_y\frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{q}\sigma}\}}{\tilde{E}_{\mathbf{q}\sigma}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta,\tilde{\mu}] \tag{3.32}$$

As we will discuss later in Sec. 3.4.2, not all these HFPs can coexist simultaneously. For now, let us move to the free energy density computation.

3.3.2 Fock contraction energy

Let us now get back to $E^{(\mathrm{AF})}_{\mathrm{F}/V}$, the double counting energy of Eq. (3.15). As said before, its definition is the same as for the normal phase, Eq. (2.10), with the averaging operation being performed on

the AF ground state:

$$E_{\mathrm{F}/V}^{(\mathrm{AF})} \equiv \frac{V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \sum_{\sigma} \left[\cos\left(\delta k_x\right) + \cos\left(\delta k_y\right)\right] \langle \hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \rangle \langle \hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \rangle \tag{3.33}$$

Proceeding exactly as in Eq. (3.16) and making the change of variables of Fig. 3.2, we get

$$E_{\mathrm{F}/V}^{(\mathrm{AF})} = \frac{V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_{\sigma} \left[\cos\left(\delta q_x\right) + \cos\left(\delta q_y\right)\right] \Big[\underbrace{\langle \hat{n}_{\mathbf{q}\sigma} \rangle \langle \hat{n}_{\mathbf{q}'\sigma} \rangle}_{\text{Diagonal terms}} - \underbrace{\langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \langle \hat{\psi}_{\mathbf{q}'\sigma} \rangle}_{\text{Off-diagonal terms}} \Big]$$

Apply the decomposition of Eq. (1.20) and use Eqns. (3.18), (3.24),

$$E_{\mathrm{F}/V}^{(\mathrm{AF})} = \frac{V}{2 L_x L_y} \sum_{\gamma,\sigma} \sum_{\mathbf{q},\mathbf{q}'} \varphi^{(\gamma)}_{\mathbf{q}} \varphi^{(\gamma)*}_{\mathbf{q}'} \left[u_{\mathbf{q}\sigma} u_{\mathbf{q}'\sigma} + v_{\mathbf{q}\sigma} v_{\mathbf{q}'\sigma}\right] \tag{3.34}$$

Now: it holds

$$u_{\mathbf{q}\sigma} = \langle \hat{n}_{\mathbf{q}\sigma} \rangle \quad \Longrightarrow \quad u_{\mathbf{q}\sigma} = u^*_{\mathbf{q}\sigma}$$

and then

$$\begin{aligned} \frac{1}{(L_x L_y)^2} \sum_{\mathbf{q},\mathbf{q}'} \varphi^{(\gamma)}_{\mathbf{q}} \varphi^{(\gamma)*}_{\mathbf{q}'} u_{\mathbf{q}\sigma} u_{\mathbf{q}'\sigma} &= \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} u_{\mathbf{q}\sigma}\right] \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}'\in\mathrm{BZ}} \varphi^{(\gamma)*}_{\mathbf{q}'} u^*_{\mathbf{q}'\sigma}\right] \\ &= \left|u^{(\gamma)}_{\sigma}\right|^2 \end{aligned}$$

At the same time,

$$v_{\mathbf{q}\sigma} = i\langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \quad \Longrightarrow \quad v_{\mathbf{q}+\boldsymbol{\pi}\sigma} = -v^*_{\mathbf{q}\sigma}$$

Since the form factors satisfy $\varphi^{(\gamma)}_{\mathbf{q}+\boldsymbol{\pi}} = -\varphi^{(\gamma)}_{\mathbf{q}}$, we get:

$$\begin{aligned} \frac{1}{(L_x L_y)^2} \sum_{\mathbf{q},\mathbf{q}'} \varphi^{(\gamma)}_{\mathbf{q}} \varphi^{(\gamma)*}_{\mathbf{q}'} v_{\mathbf{q}\sigma} v_{\mathbf{q}'\sigma} &= \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} v_{\mathbf{q}\sigma}\right] \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}'\in\mathrm{BZ}} \varphi^{(\gamma)*}_{\mathbf{q}'} v_{\mathbf{q}'\sigma}\right] \\ &= v^{(\gamma)}_{\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \varphi^{(\gamma)*}_{\mathbf{k}+\boldsymbol{\pi}} v_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right] \\ &= v^{(\gamma)}_{\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \varphi^{(\gamma)*}_{\mathbf{k}} v^*_{\mathbf{k}\sigma}\right] \\ &= v^{(\gamma)}_{\sigma} \times v^{(\gamma)*}_{\sigma} \end{aligned}$$

where in the second passage we employed the change of variables $\mathbf{q}' = \mathbf{k}+\boldsymbol{\pi}$ and used BZ periodicity to map the original sum onto a new sum for $\mathbf{k} \in \mathrm{BZ}$. Inserting the above results in Eq. (3.34), this last becomes:

$$E_{\mathrm{F}/V}^{(\mathrm{AF})} = L_x L_y \times \frac{V}{2} \sum_{\gamma,\sigma} \left[\left|u^{(\gamma)}_{\sigma}\right|^2 + \left|v^{(\gamma)}_{\sigma}\right|^2\right] \tag{3.35}$$

Interestingly enough, the new HFPs amplitudes contribute to lowering the contraction energy.

3.4 Full MFT hamiltonian and gap complexification

Finally, it is time to collect everything in a single hamiltonian operator. The MFT model is approximated by:

$$\begin{aligned} \hat{H} \simeq \sum_{\sigma} \sum_{\mathbf{k}\in\mathrm{MBZ}} \tilde{\epsilon}_{\mathbf{k}\sigma} \left[\hat{n}_{\mathbf{k}\sigma} - \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right] \\ - \sum_{\sigma} \sum_{\mathbf{k}\in\mathrm{MBZ}} \left[\tilde{\Delta}_{\mathbf{k}\sigma} \hat{\psi}_{\mathbf{k}\sigma} + \tilde{\Delta}^*_{\mathbf{k}\sigma} \hat{\psi}^{\dagger}_{\mathbf{k}\sigma}\right] - \left[E^{(\mathrm{AF})}_{\mathrm{H}/U} + E^{(\mathrm{AF})}_{\mathrm{H}/V} + E^{(\mathrm{AF})}_{\mathrm{F}/V}\right] \end{aligned}$$

where the renormalized bands are given by:

$$\tilde{\epsilon}_{\mathbf{k}\sigma} = \epsilon_{\mathbf{k}\sigma} + V \sum_{\gamma\in\{s^*,d\}} u_\sigma^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*}$$

and the full gap function by

$$\tilde{\Delta}_{\mathbf{k}\sigma} = \Delta_{\mathbf{k}\sigma} + iV \sum_{\gamma,\sigma} v_\sigma^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*}$$

The gap correction is purely imaginary, compared to the HM gap $\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} mU$. This has two obvious consequences: being no other contribution to the x pseudospin projections other than the pure HM ones, spin antisymmetry is preserved within such direction

$$\langle \hat{s}^x_{\mathbf{k}\sigma} \rangle = - \langle \hat{s}^x_{\mathbf{k}\overline{\sigma}} \rangle$$

which is relevant in the computation of the physical magnetization,

$$\begin{aligned} m &= \frac{1}{2L_xL_y} \sum_{\mathbf{r}} (-1)^{x+y} \langle \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow} \rangle \\ &= \frac{1}{2L_xL_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \left\langle \hat{s}^x_{\mathbf{k}\uparrow} - \hat{s}^x_{\mathbf{k}\downarrow} \right\rangle \\ &= \frac{1}{L_xL_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \left\langle \hat{s}^x_{\mathbf{k}\uparrow} \right\rangle \end{aligned}$$

A proof of the above derivation is given in Sec. **??**. Moreover, the net effect $\hat{H}_V$ is a "gap complexification". In other words,

$$\left| \mathrm{Im}\{\Delta_{\mathbf{k}\sigma}\} \right| \geq 0 \quad \Longrightarrow \quad \left| \langle s^y_{\mathbf{k}\sigma} \rangle \right| \geq 0$$

while for the HM the y pseudospin component was identically null.

3.4.1 The relative phase in the zero temperature half-filled ground state

In Sec. **??**, we derive analytically the AF ground state on the conventional HM at zero temperature, as a simple Bloch's transform of a polarized (tilted) pseudospins collective pure state. Eq. (**??**) reports the final form of said ground-state. Said quantum state is in general pretty similar to the conventional BCS state.

At half-filling and zero temperature, it is rather easy to extended that state to the new complexified gap. By performing an identical calculation as in Sec. **??** we here omit, it's easy to show that the new ground state for the EHM in the AF phase is identical, with RMPs in the various amplitudes and a relative phase taking care of the gap function complex phase,

$$|\Psi\rangle = \bigotimes_{\sigma} \bigotimes_{\mathbf{k}\in\mathrm{MBZ}} \left[\sin\tilde{\theta}_{\mathbf{k}\sigma} \hat{c}^\dagger_{\mathbf{k}\sigma} + e^{2i\tilde{\zeta}_{\mathbf{k}\sigma}} \cos\tilde{\theta}_{\mathbf{k}\sigma} \hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma} \right] |\Omega_{\mathbf{k}\sigma}\rangle \tag{3.36}$$

which is identical to Eq. (**??**). The angles appearing in the above Eq. (3.8), which acquire a physical meaning: [To be continued...]

3.4.2 AF phase flavours and spin degeneracy

The AF phase we are studying cannot break inversion symmetry, a core property of the system which is respected from all our approximations. This translates in the simple requirement that:

$$\tilde{E}_{\mathbf{k}\sigma} \stackrel{!}{=} \tilde{E}_{-\mathbf{k}\sigma}$$

We know that:

$$\tilde{E}_{\mathbf{k}\sigma} = \sqrt{\tilde{\epsilon}^2_{\mathbf{k}\sigma} + \mathrm{Re}^2\{\tilde{\Delta}_{\mathbf{k}\sigma}\} + \mathrm{Im}^2\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}$$

Phase	HFPs
sAF	$\{m, u^{(s^*)}, u^{(d)}, v^{(s^*)}, v^{(d)}\}$
aAF	$\{m, u^{(s^*)}, u^{(d)}, -iv^{(p_x)}, -iv^{(p_y)}\}$

Table 3.2 Set of HFP for the two possible realization of the AF phase, respectively a symmetric and an antisymmetric imaginary part of the gap.

The renormalized bare bands as well as the real part of the gap satisfy the inversion symmetry requirement,

$$\tilde{\epsilon}_{\mathbf{k}\sigma} = \tilde{\epsilon}_{-\mathbf{k}\sigma} \qquad \mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\} = \mathrm{Re}\{\tilde{\Delta}_{-\mathbf{k}\sigma}\}$$

hence inversion symmetry is respected only if

$$\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\} = \pm\,\mathrm{Im}\{\tilde{\Delta}_{-\mathbf{k}\sigma}\}$$

which is: either the imaginary part of the gap is inversion symmetric, or is antisymmetric. Based on this trivial consideration, we are able two discard two of the four self-consistency equations for the v-related HFPs: s^* and d-wave symmetries HFPs – Eqns. (3.29) and (3.30), respectively – are nonzero if $\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}$ is momentum-even; p_x and p_y-wave symmetries HFPs – Eqns. (3.31) and (3.32), respectively – are nonzero if $\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}$ is momentum-odd. The two option cannot coexist, and separate simulations can be implemented in the two symmetry sectors. We name the two possible symmetries "sAF" (symmetric antiferromagnet) and "aAF" (antisymmetric antiferromagnet). The set of HFPs then restricts in these AF structures, and is reported for both cases in Tab. 3.2.

Moreover, from now on we will omit the σ index, working just in the $\uparrow$ sector. As said in Sec. **??**, the AF phase does not break $\mathbb{Z}_2$ spin inversion symmetry, being the SDW ordering generated by a staggered but symmetric spin density distribution among sites. The MFT hamiltonian is invariant by a $\sigma \leftrightarrow \overline{\sigma}$ exchange, thus all measurable physical quantities must be. What is unmeasurable and has no effect on Physics is free to be σ-dependent, as for the sign of $\mathrm{Re}\{\Delta_{\mathbf{k}\sigma}\}$. Since all it's relevant of both the complex components of the gap function is their magnitude, we can safely treat σ as a pure degeneracy. This gives a simple equation for the last HFP,

$$\begin{aligned} m &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \langle \hat{s}^x_{\mathbf{k}\uparrow} \rangle \\ &= -\frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta, \tilde{\mu}] \end{aligned} \tag{3.37}$$

[To be continued...]

3.5 Stability of the AF phase

The discussion above is formally correct and self-consistent, thus a HF algorithm can be sketched to extract the HFPs. However we cannot proceed without considering the instability effects of Sec. **??**. As is known (and we would have liked to notice earlier) the AF-SSB at wavevector $\boldsymbol{\pi}$ we are studying here, responsible of halving the BZ down to the MBZ, can produce a stable phase on pure Hubbard lattices only at half filling [**singh1990collective**]. For the EHM the situation is even worse than that: repeating the simple Linear Reponse Theory (LRT) computations of Sec. **??** and mimicking Eq. (**??**), we get a proper response function

$$\tilde{\chi}_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\mathbf{k},\omega) = \frac{\chi_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\mathbf{k},\omega)}{1 - [U - 2V(\cos k_x + \cos k_y)]\,\chi_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\mathbf{k},\omega)} \tag{3.38}$$

[Double check the red part and conclude this paragraph...].

As is well documented in literature [**wietek2021stripes**], AF can establish in a wide range of spatial structures with particular shapes, for example stripes. In order to extract the precise AF phase, we should

1. Choose an initializer wavevector, say $\mathbf{k} = \boldsymbol{\pi}$;
2. Solve self-consistently for the HFPs at $\mathbf{k}$;
3. Compute the free energy F as a functional of the wavevector around $\mathbf{k}$;
4. Look for gradient descent of the free energy, $\boldsymbol{\nabla} F < 0$;
5. If the gradient descent is found, update $\mathbf{k} \to \mathbf{k} + \delta\mathbf{k}$ and repeat from step 2, otherwise halt.

Of course the sketched procedure can lead to a lot of complications, first the fact that at incommensurate wavevector $\mathbf{k} \neq \boldsymbol{\pi}$ the unit cell is not as simple as a pair of sites, but can become much larger requiring more refined computational solutions.

In this text we will show the self consistent result for AF-SSB at wavevector $\boldsymbol{\pi}$ even at finite doping as a first step in the above sketched procedure. The presented results are self-consistent, but unfortunately deeply unstable and unphysical at finite doping, a situation where an incommensurate and insulating AF phase is preferred to this commensurate metallic counterpart. Nevertheless we present them anyway.

3.6 AF free energy density

In order to derive the expression of the free energy density for the AF phase, we proceed precisely as in Sec. 2.4. The only notable difference is in entropy: Eq. (2.18) is written for two opposite bands on MBZ

$$s = s^{(+)} + s^{(-)}$$

with

$$s^{(\pm)} \equiv -\frac{k_B}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \Big[\Big(1 - f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) \ln \Big(1 - f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) + f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu}) \ln f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu}) \Big] \tag{3.39}$$

Consider also that we are computing free energy on the gran-canonical hamiltonian

$$\hat{H} - \mu\hat{N} \simeq \sum_{\sigma} \sum_{\mathbf{k}\in\mathrm{MBZ}} \Big[(\tilde{E}_{\mathbf{k}} - \tilde{\mu}) \hat{\gamma}^{(+)\dagger}_{\mathbf{k}\sigma} \hat{\gamma}^{(+)}_{\mathbf{k}\sigma} - (\tilde{E}_{\mathbf{k}} + \tilde{\mu}) \hat{\gamma}^{(-)\dagger}_{\mathbf{k}\sigma} \hat{\gamma}^{(-)}_{\mathbf{k}\sigma} \Big] + E_{\mathrm{c}}^{(\mathrm{AF})}$$

being $E_{\mathrm{c}}^{(\mathrm{AF})}$ the total AF contraction energy. Note that on the left-hand side we have μ and on the right-hand side its RMP counterpart $\tilde{\mu}$.

With identical meaning of the notation with respect to Sec. 2.4, the free energy contributions are:

1. Quasiparticles energy, taking care of spin degeneracy:

$$e_{\mathrm{g}}^{(\mathrm{AF})} = \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu}) - \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} + \tilde{\mu}) f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})$$

2. Contraction energy:

$$e_{\mathrm{c}}^{(\mathrm{AF})} = U(m^2 - n^2) + V \left[2zn^2 - \sum_{\gamma} \Big(\big|u^{(\gamma)}\big|^2 + \big|v^{(\gamma)}\big|^2 \Big) \right]$$

3. Entropic energy

$$e_{s}^{(\mathrm{AF})} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \ln \Big(1 - f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) - \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})$$
$$\frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \ln \Big(1 - f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) + \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} + \tilde{\mu}) f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})$$

4. Chemical potential correction

$$e_n^{(\mathrm{AF})} = \mu n$$

Compose everything

$$f_{\mathrm{MFT}}^{(\mathrm{AF})} \equiv e_{\mathrm{g}}^{(\mathrm{AF})} + e_{\mathrm{c}}^{(\mathrm{AF})} + e_s^{(\mathrm{AF})} + e_n^{(\mathrm{AF})}$$

and we get

$$f_{\mathrm{MFT}}^{(\mathrm{AF})} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \left[\ln\left(1 - f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\right) + \ln\left(1 - f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\right)\right] + U(m^2 - n^2) + V\left[2zn^2 - \sum_{\gamma}\left(\left|u^{(\gamma)}\right|^2 + \left|v^{(\gamma)}\right|^2\right)\right] + \mu n \quad (3.40)$$

It's of particular interest the half-filling condition, with $\tilde{\mu} = 0$. [To be continued...]

3.7 Numerical simulations

[To be continued...]

3.7.1 HF iterative algorithm sketch

[To be continued...]

Chapter 4

Superconducting instability

This chapter is devoted to studying the superconducting phase of the system. The only symmetry we assume to break is the $\mathrm{U^c}(1)$ charge symmetry, thus allowing for superconducting fluctuations. As is described thoroughly in Sec. 4.3.1, the hopping amplitude is renormalized because of the non-local attraction. The symmetry structure of the pairing mechanism determines the contributing Cooper fluctuations: for s-wave and d-wave superconductivity, only the o.s. Cooper term contributes; for p_ℓ-wave superconductivity, the s.s. term contributes as well. In the following sections, a derivation containing both Cooper terms is proposed.

4.1 The SC phase and its symmetries

The starting point is, as done for the other phases, the discussion of the symmetries of the SC phase. We said in introduction we are studying the

$$\mathrm{SU}(2) \times \mathrm{U^c}(1) \times (\text{Point group}) \quad \overset{\text{SSB}}{\to} \quad \mathrm{SU}(2) \times (\text{Point group})$$

symmetry breaking. Therefore the system density we assume to be uniform across all sites, similarly to the normal phase discussed in Chap. 2. [To be continued...]

4.1.1 Superconductivity in the HM

In Sec. **??**, we discuss in detail the (impossibility of a stable) s-wave SC phase in the conventional HM. This is an obvious result, since the two-body interaction thereby present $\hat{H}_U$ is local and repulsive. We nevertheless treat the model within the MFT framework, and here recall the main widely known results, and later use them as groundwork for the EHM extension. Similarly to Sec. 3.1.1, let us define the operators

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} \tag{4.1}$$

$$\hat{\phi}_{\mathbf{k}} \equiv \hat{c}^\dagger_{\mathbf{k}\uparrow}\hat{c}^\dagger_{-\mathbf{k}\downarrow} \tag{4.2}$$

whose EVs are relevant for the SC phase. Let:

$$u_{\mathbf{q}\sigma} \equiv \langle \hat{n}_{\mathbf{q}\sigma} \rangle \tag{4.3}$$

$$w_{\mathbf{q}\sigma} \equiv \langle \hat{\phi}^\dagger_{\mathbf{q}} \rangle \tag{4.4}$$

The MFT hamiltonian is given explicitly for the HM in Eq. (**??**),

$$\hat{H}_t + \hat{H}_U \overset{\text{MFT}}{\simeq} \sum_{\mathbf{k}\in\mathrm{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}\sigma} \left[\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} - \hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right] - \sum_{\mathbf{q}\in\mathrm{BZ}} \left[\Delta^{(s)}\hat{\phi}_{\mathbf{q}} + \Delta^{(s)*}\hat{\phi}^\dagger_{\mathbf{q}}\right] + nU \times \hat{N} - \left[E^{(\mathrm{SC})}_{\mathrm{H}/U} + E^{(\mathrm{SC})}_{\mathrm{C}/U}\right]$$

EV	Meaning	Equation
$\langle \hat{s}^x_{\mathbf{k}} \rangle$	$\left\langle \hat{\phi}_{\mathbf{k}} + \hat{\phi}^\dagger_{\mathbf{k}} \right\rangle$	(**??**)
$i \langle \hat{s}^y_{\mathbf{k}} \rangle$	$\left\langle \hat{\phi}_{\mathbf{k}\sigma} - \hat{\phi}^\dagger_{\mathbf{k}} \right\rangle$	(**??**)
$\langle \hat{s}^z_{\mathbf{k}} \rangle$	$\langle \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{\mathbf{k}\downarrow} \rangle - 1$	(**??**)

Table 4.1 List of the various expectation values (EVs) of each pseudospin component, their meaning in terms of fermionic operators and the reference equations from the pure HM. The z component refers also to Eq. (**??**).

with $E^{(\mathrm{SC})}_{\mathrm{H}/U}$, $E^{(\mathrm{SC})}_{\mathrm{C}/U}$ the contraction energies arising from Hartree and Cooper channels, respectively in Eq. (**??**) and (**??**). In deriving this result we decomposed the interaction of the intermediate passage of Eq. (**??**) and assumed the Cooper pair to have null total momentum. Moreover, we recognized in the passage of Eq. (**??**) the presence of a s-wave harmonic transform of the function $w_{\mathbf{q}}$, and thus defined in Eq. (**??**),

$$w^{(s)} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{q}\in \mathrm{BZ}} w_{\mathbf{q}} \quad \text{and} \quad \Delta^{(s)} \equiv -U w^{(s)}$$

Nambu-Bogoliubov representation of the MFT hamiltonian is given in Eq. (**??**) and equivalently in Eq. (**??**),

$$\hat{H}_t + \hat{H}_U - \mu \hat{N} \simeq \sum_{\mathbf{k}\in \mathrm{BZ}} \sum_{\sigma} \hat{\Phi}^\dagger_{\mathbf{k}} h_{\mathbf{k}} \hat{\Phi}_{\mathbf{k}} - \left[E^{(\mathrm{SC})}_{\mathrm{A}/t} + E^{(\mathrm{SC})}_{\mathrm{H}/U} + E^{(\mathrm{SC})}_{\mathrm{C}/U} \right]$$

being

$$h_{\mathbf{k}} \equiv \begin{bmatrix} \xi_{\mathbf{k}} & -\Delta_{\mathbf{k}} \\ -\Delta^*_{\mathbf{k}} & -\xi_{\mathbf{k}} \end{bmatrix} \quad \text{and} \quad \xi_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} - \tilde{\mu}$$

and $\Delta_{\mathbf{k}} = \Delta^{(s)}$. Nambu spinors are, in this case,

$$\hat{\Phi}_{\mathbf{k}} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\uparrow} \\ \hat{c}^\dagger_{-\mathbf{k}\downarrow} \end{bmatrix}$$

For the SC phase in the HM model, in general, we can admit for a complex gap function – although being the gap a constant it can always be reduced to a purely real gap. Diagonalization by the angles defined in Eq. (**??**) is performed via the composed rotation of Eq. (**??**),

$$W_{\mathbf{k}} = \begin{bmatrix} \cos\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & -\sin\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \\ \sin\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & \cos\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \end{bmatrix}$$

The Bogoliubov bands are $\pm E_{\mathbf{k}} = \sqrt{\xi^2_{\mathbf{k}} + |\Delta_{\mathbf{k}}|^2}$. The only crucial difference with the AF Bogoliubov bands it that here, apart from the absence of the spin index – a degeneracy that is solved now by enlarging the computation to the entire BZ –, also the bare bands $\epsilon_{\mathbf{k}}$ are substituted by their shifted version $\xi_{\mathbf{k}}$. This difference is essential: it is this feature that makes the superconducting phase gapped in an insulating sense, which means, opening the gap at the position of $\tilde{\mu}$. Instead for the AF phase the gap was always opened at "midband", the MBZ contour where $\epsilon_{\mathbf{k}} = 0$. As a final remark, as is proven trivially at the end of Sec. **??**, s-wave SC is impossible at MFT level on the pure HM. This was obvious, due to the necessity of an even small attractive interaction to form Cooper pairs. To conclude, in Tab. 4.1 we report the operator-pseudospin map for each direction.

4.1.2 Extension to the EHM

Add now the non-local interaction $\hat{H}_V$: translational invariance is preserved while charge conservation is broken, thus all three channels (Hartree, Fock and Cooper) contribute via RWCs to the MFT hamiltonian. As for the AF phase, we aim to derive a MFT solution for the SC phase which

is an extension of the HM solution of Sec. **??**, with an effective redefinition of the various physical quantities and mathematical objects,

$$\xi_{\mathbf{k}} \to \tilde{\xi}_{\mathbf{k}} \qquad E_{\mathbf{k}} \to \tilde{E}_{\mathbf{k}} \qquad \Delta_{\mathbf{k}} \to \tilde{\Delta}_{\mathbf{k}} \qquad \mu \to \tilde{\mu}$$

To be completely correct, the chemical potential was already renormalized in the HM derivation in Sec. **??**. It here acquires yet another renormalization. The band energies renormalization is simply

$$\tilde{E}_{\mathbf{k}} \equiv \sqrt{\tilde{\xi}_{\mathbf{k}}^2 + |\tilde{\Delta}_{\mathbf{k}}|^2}$$

and as for the AF phase:

$$\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \tau^x \hat{\Psi}_{\mathbf{k}\sigma} \rangle = -\sin(2\tilde{\theta}_{\mathbf{k}\sigma}) \sin(2\tilde{\zeta}_{\mathbf{k}\sigma}) \langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \tag{4.5}$$

$$\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \tau^y \hat{\Psi}_{\mathbf{k}\sigma} \rangle = -\sin(2\tilde{\theta}_{\mathbf{k}\sigma}) \cos(2\tilde{\zeta}_{\mathbf{k}\sigma}) \langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \tag{4.6}$$

$$\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Psi}_{\mathbf{k}\sigma} \rangle = \cos(2\tilde{\theta}_{\mathbf{k}\sigma}) \langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \tag{4.7}$$

These equations are identical to the ones for the pure HM, Eqns. (**??**), (**??**) and (**??**). In the tilted frame, we have the analogous of Eq. (3.6) with the important exception that the chemical potential in the Fermi-Dirac factors is zero,

$$\langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle = \frac{1}{2} \left[f\left(\tilde{E}_{\mathbf{k}\sigma}; \beta, 0\right) - f\left(-\tilde{E}_{\mathbf{k}\sigma}; \beta, 0\right) \right] \tag{4.8}$$

This is because all the chemical potential operatorial contributions were absorbed in the bare bands $\tilde{\xi}_{\mathbf{k}}$, thus the pseudospin state is thermally described by an effective null chemical potential.

The (renormalized) thermal factors as in Eq. (**??**) are

$$\tilde{\mathrm{Th}}^{(\pm)}_{\mathbf{k}\sigma}[\beta, \tilde{\mu}] \equiv f\left(\tilde{E}_{\mathbf{k}\sigma}; \beta, 0\right) \pm f\left(-\tilde{E}_{\mathbf{k}\sigma}; \beta, 0\right) \tag{4.9}$$

while for the angles it holds

$$\sin(2\tilde{\xi}_{\mathbf{k}\sigma}) = \frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{|\tilde{\Delta}_{\mathbf{k}\sigma}|} \quad \cos(2\tilde{\xi}_{\mathbf{k}\sigma}) = \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{|\tilde{\Delta}_{\mathbf{k}\sigma}|} \quad \sin(2\tilde{\theta}_{\mathbf{k}\sigma}) = \frac{|\tilde{\Delta}_{\mathbf{k}\sigma}|}{\tilde{E}_{\mathbf{k}\sigma}} \quad \cos(2\tilde{\theta}_{\mathbf{k}\sigma}) = \frac{\tilde{\xi}_{\mathbf{k}\sigma}}{\tilde{E}_{\mathbf{k}\sigma}} \tag{4.10}$$

From Tabs. 1.2 and 1.3, we know all channel contribute. It is already known the effect of the Fock contractions, responsible as for the normal phase (Chap. 2) of the bare bands renormalization. Cooper terms are to be handled more carefully, since can contribute to superconductivity both in the s.s. and in the o.s. spin sectors.

4.2 Non-local Hartree effects: chemical potential renormalization

As always, let us start with a very brief section about the Hartree effects on the MFT decomposition of the hamiltonian. We will not delve in the detailed calculations of the Hartree channel MFT effects, already described in previous chapter as well as in Sec. **??**. The only effect here is the chemical potential renormalization.

4.2.1 Hartree channel in the superconducting phase: analytical derivation

The Hartree fluctuations are density-density couplings in the MFT hamiltonian. Knowing the deal with a uniform-density phase, we can re-use as it is the derivation of Sec. 2.2.1. Without repeating the calculations, the $\hat{H}_V$ is reduced in the Hartree channel to

$$\hat{H}_V \simeq -2nzV \times \hat{N} - E^{(\mathrm{SC})}_{\mathrm{H}/V} + (\text{all the rest})$$

This, together with the $\hat{H}_U$ Hartree channel decomposition

$$\hat{H}_U \simeq nU \times \hat{N} - E^{(\mathrm{SC})}_{\mathrm{H}/U}$$

gives the chemical potential full renormalization, identical to Eq. (2.5),

$$\tilde{\mu} \equiv \mu + n(2zV - U) \tag{4.11}$$

4.2.2 Hartree contraction energy

The contraction energy for the non-local sector was derived in Eq. (2.7),

$$E^{(\mathrm{SC})}_{\mathrm{H}/V} = -L_x L_y \times 2zn^2 V$$

and together with the local counterpart of Eq. (2.6)

$$E^{(\mathrm{SC})}_{\mathrm{H}/U} = L_x L_y \times n^2 U$$

Now we move to the more interesing Fock sector.

4.3 Fock effects: bare bands renormalization

[To be continued...]

4.3.1 Fock channel in the superconducting phase: analytical derivation

When it comes to the Fock RWCs, being the SC phase density-uniform we can re-use almost the entirety of the derivation of Sec. 2.3. Setting as in Eq. (2.14)

$$u_{\mathbf{q}\sigma} \equiv \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle \tag{4.12}$$

Eq. (2.15) stands perfectly valid, translated to the SC phase

$$\begin{aligned} V \sum_{\langle ij \rangle} \sum_{\sigma} \Big[\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle \Big] \\ \stackrel{\mathcal{F}}{=} \sum_{\gamma,\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q} \in \mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} u_{\mathbf{q}\sigma} \right] \times V \sum_{\mathbf{q}' \in \mathrm{BZ}} \varphi^{(\gamma)*}_{\mathbf{q}'} \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} - E^{(\mathrm{SC})}_{\mathrm{F}/V} \end{aligned} \tag{4.13}$$

being

$$E^{(\mathrm{SC})}_{\mathrm{F}/V} \equiv \frac{V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_{\sigma} \left[\cos(\delta q_x) + \cos(\delta q_y) \right] \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle \langle \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} \rangle \tag{4.14}$$

To get to this point, we employed the assumption

$$\langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}'\sigma'} \rangle = \delta_{\mathbf{k}=\mathbf{k}'} \delta_{\sigma=\sigma'} \langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \rangle$$

which essentially states translational invariance and spin degeneracy. This assumption is obviously well-verified by the BCS ground state of Eq. [Missing]. Recognising the symmetry decomposition

$$u^{(\gamma)}_{\sigma} = \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle$$

due to inversion symmetry the conclusion is identical to Eqns. (2.15) and (2.16),

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv \epsilon_{\mathbf{k}\sigma} + u^{(s^*)}_{\sigma} V (\cos q_x + \cos q_y) + u^{(d)}_{\sigma} V (\cos q_x - \cos q_y)$$

In order to estimate the γ-wave components renormalizing the bare bands, we need to compute separately in spin sectors the EVs of $u_{\mathbf{k}\sigma}$

$$
\begin{aligned}
u_{\mathbf{k}\uparrow} &= \langle \hat{c}^\dagger_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow}\rangle \qquad (4.15)\\
&= \left\langle \hat{\Phi}^\dagger_{\mathbf{k}} \frac{\mathbb{I}+\tau^z}{2}\hat{\Phi}_{\mathbf{k}}\right\rangle \\
&= \frac{1}{2}\left[\left\langle \hat{\Gamma}^\dagger_{\mathbf{k}}\hat{\Gamma}_{\mathbf{k}}\right\rangle + \langle \hat{s}^z_{\mathbf{k}}\rangle\right] \\
&= \frac{1}{2}\left[\tilde{\mathrm{Th}}^{(+)}_{\mathbf{k}}[\beta] + \cos 2\tilde{\theta}_{\mathbf{k}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta]\right] \\
&= \frac{1+\cos 2\tilde{\theta}_{\mathbf{k}}}{2} f\left(\tilde{E}_{\mathbf{k}};\beta,0\right) + \frac{1-\cos 2\tilde{\theta}_{\mathbf{k}}}{2} f\left(-\tilde{E}_{\mathbf{k}};\beta,0\right) \\
&= \cos^2\tilde{\theta}_{\mathbf{k}} f\left(\tilde{E}_{\mathbf{k}};\beta,0\right) + \sin^2\tilde{\theta}_{\mathbf{k}} f\left(-\tilde{E}_{\mathbf{k}};\beta,0\right) \\
&= \sin^2\tilde{\theta}_{\mathbf{k}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right) + f\left(\tilde{E}_{\mathbf{k}};\beta,0\right) \qquad (4.16)
\end{aligned}
$$

having we used the simple relation

$$
\begin{aligned}
\frac{1}{e^{-x}+1} - \frac{1}{e^{x}+1} &= \frac{e^x-1}{e^x+1} \\
&= \tanh\left(\frac{x}{2}\right)
\end{aligned}
$$

For the spin $\downarrow$ EV,

$$
\begin{aligned}
u_{-\mathbf{k}\downarrow} &= \langle \hat{c}^\dagger_{-\mathbf{k}\downarrow}\hat{c}_{-\mathbf{k}\downarrow}\rangle \\
&= 1 - \langle \hat{c}_{-\mathbf{k}\downarrow}\hat{c}^\dagger_{-\mathbf{k}\downarrow}\rangle \\
&= 1 - \left\langle \hat{\Phi}^\dagger_{\mathbf{k}} \frac{\mathbb{I}-\tau^z}{2}\hat{\Phi}_{\mathbf{k}}\right\rangle \\
&= 1 - \frac{1}{2}\left[\left\langle \hat{\Gamma}^\dagger_{\mathbf{k}}\hat{\Gamma}_{\mathbf{k}}\right\rangle - \langle \hat{s}^z_{\mathbf{k}}\rangle\right] \\
&= 1 - \frac{1}{2}\left[\tilde{\mathrm{Th}}^{(+)}_{\mathbf{k}}[\beta] - \cos 2\tilde{\theta}_{\mathbf{k}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta]\right] \\
&= 1 - \frac{1-\cos 2\tilde{\theta}_{\mathbf{k}}}{2} f\left(\tilde{E}_{\mathbf{k}};\beta,0\right) - \frac{1+\cos 2\tilde{\theta}_{\mathbf{k}}}{2} f\left(-\tilde{E}_{\mathbf{k}};\beta,0\right) \\
&= 1 - \sin^2\tilde{\theta}_{\mathbf{k}} f\left(\tilde{E}_{\mathbf{k}};\beta,0\right) - \cos^2\tilde{\theta}_{\mathbf{k}} f\left(-\tilde{E}_{\mathbf{k}};\beta,0\right) \\
&= 1 - \sin^2\tilde{\theta}_{\mathbf{k}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right) - f\left(-\tilde{E}_{\mathbf{k}};\beta,0\right) \\
&= -\sin^2\tilde{\theta}_{\mathbf{k}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right) + f\left(\tilde{E}_{\mathbf{k}};\beta,0\right) \qquad (4.17)
\end{aligned}
$$

These results will be used to extract self-consistency equations for each HFP. Note that

$$
\lim_{|\Delta_{\mathbf{k}}|\to 0} u_{\mathbf{k}\sigma} = f(\tilde{\xi}_{\mathbf{k}};\beta,0)
$$

Correctly, the null-gap limit reduces to a perfectly degenerate Fermi gas.

4.3.2 Fock contraction energy

In terms of the energy shift due to Wick's contraction, Eq. (2.17) holds identically

$$
E^{(\mathrm{SC})}_{\mathrm{F}/V} = L_x L_y \times \frac{V}{2}\sum_\sigma \left[\left|u^{(s^*)}\right|^2 + \left|u^{(d)}\right|^2\right]
$$

with the only obvious difference that the $u^{(\gamma)}$ are computed on the SC ground state by the means of Eqns. (4.16), (4.17).

Spatial structure	Pairing channel	Relevant pairing
Symmetric wave function	Singlet pairing	Just $\langle \hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\bar{\sigma}}\rangle$
Anti-symmetric wave function	Triplet pairing	Both $\langle \hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\bar{\sigma}}\rangle$ and $\langle \hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\sigma}\rangle$

Table 4.2 Relation of the Cooper pairing channel with the wavefunction spatial symmetry (intended as the inversion $(x, y) \to (-x, -y)$).

4.4 Cooper effects in the singlet channel: gap renormalization

Let us start once again from the general EHM of Eq. (**??**),

$$\hat{H} = \underbrace{-t\sum_{\langle ij\rangle}\sum_{\sigma}\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}}_{\hat{H}_t} \underbrace{+U\sum_{i}\hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{i\downarrow}\hat{c}_{i\downarrow}\hat{c}_{i\uparrow}}_{\hat{H}_U} \underbrace{-V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\sigma'}\hat{c}_{j\sigma'}\hat{c}_{i\sigma}}_{\hat{H}_V}$$

As is discussed in Sec. 1.3, when applying Wick's theorem the resulting terms break the natural symmetries of the model. The superconducting symmetry we study breaks just the $\mathrm{U}^{\mathrm{c}}(1)$ charge conservation. Now, when dealing with superconducting Cooper pairing we need to account also for the spatial structure of the Cooper pair itself. Consider the generic Cooper fluctuation

$$\langle \hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\sigma'}\rangle \quad \text{with } i, j \text{ NN}$$

From basic Quantum Mechanics we know the summation rules of the spin algebra $\mathfrak{su}(2)$,

$$\frac{1}{2}\otimes\frac{1}{2} = 0 \oplus 1$$

The two pairing channels are, at this level, the singlet channel associated to total spin 0 and the triplet channel associated to total spin 1. If we impose a specific spatial symmetry on the hamiltonian the ground state wavefunction will follow naturally, and the pairing channel will be the one providing a total anti-symmetry to the full wavefunction. This gives a selection rule over the relevant pairings: if we work with space symmetric structures – say, s^*-wave or d-wave – the pairing will happen in singlet channel, allowing us to eliminate the triplet pairing. This concept is summarized in Tab. 4.2.

4.5 Cooper fluctuations in the opposite-spin sector

This section deals with Cooper fluctuations induced by the o.s. part of the non-local hamiltonian –referring to the notation of Eq. (1.7)– somewhat the simplest form of Cooper pairing. This sector contributes both to singlet pairing and triplet pairing. Considering Cooper fluctuations in the singlet channel, we need to break $\mathrm{U}^{\mathrm{c}}(1)$ symmetry imposing space inversion symmetry, while for the triplet anti-symmetry is required. Let us now break down the MFT discussion for the local and non-local interactions.

Local interaction U. The detailed treatment of $\hat{H}_U$ is discussed in Sec. **??**, in the context of conventional HM superconductivity. We hereby just sketch the most important results. First, similarly to the AF operator of Eq. (3.2) let the Cooper fluctuations operator be

$$\hat{\phi}_{\mathbf{k}} \equiv \hat{c}_{-\mathbf{k}\downarrow}\hat{c}_{\mathbf{k}\uparrow} \tag{4.18}$$

Consider first the local part,

$$\hat{H}_U = U\sum_{i\in\mathcal{S}}\hat{n}_{i\uparrow}\hat{n}_{i\downarrow} \simeq U\sum_{i\in\mathcal{S}}\left[\langle \hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{i\downarrow}\rangle\hat{c}_{i\downarrow}\hat{c}_{i\uparrow} + \hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{i\downarrow}\langle \hat{c}_{i\downarrow}\hat{c}_{i\uparrow}\rangle\right]$$

and use the result of Eq. (1.15),

$$\hat{H}_U \simeq \frac{U}{L_xL_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow}\rangle\hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} + \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow}\langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow}\rangle\right]$$

We are not breaking translational invariance, thus only Cooper fluctuations with net zero total momentum are allowed. This means only $\mathbf{K} = \mathbf{0}$ contributes. Define the pairing operators

$$\hat{\phi}_{\mathbf{k}} \equiv \hat{c}_{-\mathbf{k}\downarrow}\hat{c}_{\mathbf{k}\uparrow} \qquad \hat{\phi}^\dagger_{\mathbf{k}} \equiv \hat{c}^\dagger_{\mathbf{k}\uparrow}\hat{c}^\dagger_{-\mathbf{k}\downarrow}$$

Then the non local repulsion reduces to the ordinary BCS-like interaction,

$$\hat{H}_U \simeq \sum_{\mathbf{k}} \left[\mathcal{U}_{\mathbf{k}}\hat{\phi}_{\mathbf{k}} + \mathcal{U}^*_{\mathbf{k}}\hat{\phi}^\dagger_{\mathbf{k}}\right] \tag{4.19}$$

where the MFT parameter $\mathcal{U}_{\mathbf{k}}$ must satisfy the self-consistency equation

$$\mathcal{U}_{\mathbf{k}} \equiv \frac{U}{L_x L_y}\sum_{\mathbf{k}} \langle \hat{\phi}^\dagger_{\mathbf{k}}\rangle \tag{4.20}$$

Note that $\mathcal{U}_{\mathbf{k}}$ is actually momentum independent. This is due to the fact that the repulsion is completely localized.

Non-local interaction V. The non-local attraction in the opposite-spin sector of $\hat{H}_V$ is given by

$$\hat{H}_V^{(\text{o.s.})} = -V\sum_{\langle ij\rangle}\sum_{\sigma}\hat{n}_{i\sigma}\hat{n}_{j\overline{\sigma}}$$

which, using Eq. (1.9) and performing Wick's contractions, reduces to:

$$\begin{aligned}\hat{H}_V^{(\text{o.s.})} &= -V\sum_{i\in\mathcal{S}}\sum_{\ell=x,y}\sum_{\delta=\pm\delta_\ell}\hat{n}_{i\uparrow}\hat{n}_{i+\delta\downarrow}\\ &\simeq -V\sum_{i\in\mathcal{S}}\sum_{\ell=x,y}\sum_{\delta=\pm\delta_\ell}\left[\langle\hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{i+\delta\downarrow}\rangle\hat{c}_{i+\delta\downarrow}\hat{c}_{i\uparrow} + \hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{i+\delta\downarrow}\langle\hat{c}_{i+\delta\downarrow}\hat{c}_{i\uparrow}\rangle\right]\end{aligned}$$

Using Eq. (1.11) we can move to reciprocal space,

$$\begin{aligned}\hat{H}_V^{(\text{o.s.})} \simeq -\frac{V}{L_x L_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}&\left[\cos(\delta k_x) + \cos(\delta k_y)\right]\\ &\times\left[\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow}\rangle\hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} + \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow}\langle\hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow}\rangle\right]\end{aligned}$$

Identical considerations as above hold, and just the $\mathbf{K} = \mathbf{0}$ term contributes. We have finally

$$\hat{H}_V^{(\text{o.s.})} \simeq -\sum_{\mathbf{k},\mathbf{k}'} V_{\mathbf{k}\mathbf{k}'}\left[\langle\hat{\phi}^\dagger_{\mathbf{k}}\rangle\hat{\phi}_{\mathbf{k}'} + \langle\hat{\phi}_{\mathbf{k}}\rangle\hat{\phi}^\dagger_{\mathbf{k}'}\right]$$

where the two-body potential was defined

$$V_{\mathbf{k}\mathbf{k}'} = \frac{V}{L_x L_y}\left[\cos(\delta k_x) + \cos(\delta k_y)\right]$$

Making use of the decomposition of Eq. (1.20), the two-body potential becomes

$$\begin{aligned}V_{\mathbf{k}\mathbf{k}'} &= \frac{V}{2L_x L_y}\sum_{\gamma}\varphi^{(\gamma)}_{\mathbf{k}}\varphi^{(\gamma)*}_{\mathbf{k}'}\\ &= \sum_{\gamma} V^{(\gamma)}\varphi^{(\gamma)}_{\mathbf{k}}\varphi^{(\gamma)*}_{\mathbf{k}'}\end{aligned}$$

being $\gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\}$ and $\varphi^{(\gamma)}_{\mathbf{k}}$ the reciprocal-space expressions for the form factors of Tab. 1.4, listed explicitly in Tab. 1.5, and $V^{(\gamma)}_{\mathbf{k}\mathbf{k}'}$ the symmetry-resolved components of the non-local attraction. Then the two-body potential has been decomposed in its planar symmetry components, each of which will naturally couple only to identically structured parameters in the full hamiltonian.

Define now the non-local gap function

$$\mathcal{V}_{\mathbf{k}} \equiv \sum_{\mathbf{k}'} V_{\mathbf{k}\mathbf{k}'} \langle \hat{\phi}^\dagger_{\mathbf{k}'} \rangle \tag{4.21}$$

one gets immediately

$$\hat{H}_V \simeq -\sum_{\mathbf{k}} \left[\mathcal{V}_{\mathbf{k}} \hat{\phi}_{\mathbf{k}} + \mathcal{V}^*_{\mathbf{k}} \hat{\phi}^\dagger_{\mathbf{k}} \right] \tag{4.22}$$

To assume symmetry is broken in a specific symmetry channel γ means precisely to assume $\langle \hat{\phi}_{\mathbf{k}} \rangle \propto \varphi^{(\gamma)}_{\mathbf{k}}$. Of course, in Eq. (4.21) only the γ component of the potential survives, implying the gap function acquires the same symmetry,

$$\mathcal{V}_{\mathbf{k}} \propto \sum_{\mathbf{k}'} \varphi^{(\gamma)}_{\mathbf{k}} \varphi^{(\gamma)*}_{\mathbf{k}'} \varphi^{(\gamma)}_{\mathbf{k}'} \propto \varphi^{(\gamma)}_{\mathbf{k}}$$

where orthonormality of the $\varphi^{(\gamma)}_{\mathbf{k}}$ functions of Tab. 1.5 was used. Thus, assuming to have superconductivity in a given sector γ, the self consistency equation reads

$$\forall \gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\} \qquad \mathcal{V}_{\mathbf{k}}\big|_\gamma \equiv \frac{V}{2L_x L_y} \sum_{\mathbf{k}} \varphi^{(\gamma)}_{\mathbf{k}} \langle \hat{\phi}^\dagger_{\mathbf{k}} \rangle \tag{4.23}$$

Now we merge the two interaction in a single gap function.

4.5.1 Full gap function and self-consistency equations

Define the full gap function as the sum of both contributions,

$$\Delta_{\mathbf{k}} \equiv \mathcal{V}_{\mathbf{k}} - \mathcal{U}_{\mathbf{k}} \tag{4.24}$$

The full self-consistency equation is given by the simple combination of Eqns. (4.20) and (4.23),

$$\Delta_{\mathbf{k}} \equiv \sum_{\mathbf{k}'} \left[V^{(s)} + V_{\mathbf{k}\mathbf{k}'} \right] \langle \hat{\phi}^\dagger_{\mathbf{k}'} \rangle \quad \text{with} \quad V^{(s)} = -\frac{U}{2L_x L_y} \tag{4.25}$$

The gap function decomposes in symmetry channels as well,

$$\Delta_{\mathbf{k}} = \sum_{\gamma} \Delta^{(\gamma)} \varphi^{(\gamma)}_{\mathbf{k}}$$

If SC arises in a specific symmetry channel, $\Delta_{\mathbf{k}}$ will show the same symmetry. It follows, due to orthonormality and using Eq. (4.25),

$$\begin{aligned}
\Delta^{(\gamma)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k}} \varphi^{(\gamma)*}_{\mathbf{k}} \Delta_{\mathbf{k}} \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k}} \varphi^{(\gamma)*}_{\mathbf{k}} \sum_{\mathbf{k}'} \left[V^{(s)} + V_{\mathbf{k}\mathbf{k}'} \right] \langle \hat{\phi}^\dagger_{\mathbf{k}'} \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k}} \varphi^{(\gamma)*}_{\mathbf{k}} \sum_{\mathbf{k}'\gamma'} V^{(\gamma')} \varphi^{(\gamma')}_{\mathbf{k}} \varphi^{(\gamma')*}_{\mathbf{k}'} \langle \hat{\phi}^\dagger_{\mathbf{k}'} \rangle \\
&= V^{(\gamma)} \sum_{\mathbf{k}} \varphi^{(\gamma)*}_{\mathbf{k}} \langle \hat{\phi}^\dagger_{\mathbf{k}} \rangle
\end{aligned} \tag{4.26}$$

This result provides a set of self-consistency equations for each symmetry channel, listed in Tab. 4.3. Notice that to reconstruct self-consistently the full s-wave phase transition, the actual gap function is given by

$$\Delta^{(s)} + \Delta^{(s^*)}(c_x + c_y)$$

The s-wave transition is the only one equipped of both the local and the non-local parts.

Structure	Self-consistency equation	Graph
s-wave	$\Delta^{(s)} = -\frac{U}{2L_xL_y}\sum_{\mathbf{k}}\langle\hat{\phi}^\dagger_{\mathbf{k}}\rangle$	Fig. 1.4a
Extended s-wave	$\Delta^{(s^*)} = \frac{V}{L_xL_y}\sum_{\mathbf{k}}(c_x+c_y)\langle\hat{\phi}^\dagger_{\mathbf{k}}\rangle$	Fig. 1.4b
p_x-wave	$\Delta^{(p_x)} = -i\sqrt{2}\frac{V}{L_xL_y}\sum_{\mathbf{k}}s_x\langle\hat{\phi}^\dagger_{\mathbf{k}}\rangle$	Fig. 1.4c
p_y-wave	$\Delta^{(p_y)} = -i\sqrt{2}\frac{V}{L_xL_y}\sum_{\mathbf{k}}s_y\langle\hat{\phi}^\dagger_{\mathbf{k}}\rangle$	Fig. 1.4d
$d_{x^2-y^2}$-wave	$\Delta^{(d)} = \frac{V}{L_xL_y}\sum_{\mathbf{k}}(c_x-c_y)\langle\hat{\phi}^\dagger_{\mathbf{k}}\rangle$	Fig. 1.4e

Table 4.3 Symmetry resolved self-consistency equations for the MFT parameters $\Delta^{(\gamma)}$, based on Eq. (4.25) and (4.26). By computing $\langle\hat{\phi}^\dagger_{\mathbf{k}}\rangle$, it is possible to reconstruct the various components of the gap function.

4.5.2 Non-local bands renormalization in the same-spin sector

In the context of antiferromagnetism (Sec. **??**) we discussed the role of the non-local interaction $\hat{H}_V$ as a source of bands effective renormalization. One of the most interesting results was the on of Eq. (**??**), with the hopping parameter being rigidly shifted. In the present context, we aim to treat the non-local part similarly by using immediately the aforementioned results. Start from Eq. (**??**),

$$V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}+\text{h.c.}\right]$$
$$=\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\sum_{\sigma}\left[\cos(\delta k_x)+\cos(\delta k_y)\right]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}$$

For the BCS ground state, it is immediate to see the only relevant contribution comes from

$$\mathbf{k}=-\mathbf{k}'$$

which gives the diagonal part of Eq. (**??**). We are then left with (half) the result of Eq. (**??**)

$$V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}+\text{h.c.}\right]=\frac{2V}{L_xL_y}\sum_{\mathbf{q},\mathbf{q}'}\sum_{\sigma}\left[\cos(\delta q_x)+\cos(\delta q_y)\right]\langle\hat{c}^\dagger_{\mathbf{q}\sigma}\hat{c}_{\mathbf{q}\sigma}\rangle\hat{c}^\dagger_{\mathbf{q}'\sigma}\hat{c}_{\mathbf{q}'\sigma}\tag{4.27}$$

Recall Eq. (1.20): the structure term decomposes in harmonic waves, and this feature is particularly handy in order to decouple the $\mathbf{q}$ and $\mathbf{q}'$ parts, as we later do.

4.6 Superconducting solution in the Nambu formalism

As a first approach to anisotropic SC in the Hubbard model, let us discuss the superconducting solutions in the simple scenario where crystal translational invariance is preserved and thus the system can be treated by the means of common BCS. As will be discussed, the non-local attraction acts as a source of SC in all symmetry sectors.

Define the Nambu spinor[1] as in BCS

$$\hat{\Psi}_{\mathbf{k}}\equiv\begin{bmatrix}\hat{c}_{\mathbf{k}\uparrow}\\ \hat{c}^\dagger_{-\mathbf{k}\downarrow}\end{bmatrix}$$

[1]Notice that the spinor is here differently defined with respect to App. **??**, where because of the HF prevalence in mean-field decoupling the spinor components were homogeneously fermions creations or destructions.

Evidently,

$$\phi_{\mathbf{k}} = \hat{\Psi}^\dagger_{\mathbf{k}} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \hat{\Psi}_{\mathbf{k}} \qquad \phi^\dagger_{\mathbf{k}} = \hat{\Psi}^\dagger_{\mathbf{k}} \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \hat{\Psi}_{\mathbf{k}} \tag{4.28}$$

The full hamiltonian is then given by:

$$\hat{H} = \sum_{\mathbf{k}} \hat{\Psi}_{\mathbf{k}} h_{\mathbf{k}} \hat{\Psi}_{\mathbf{k}} \qquad h_{\mathbf{k}} \equiv \begin{bmatrix} \epsilon_{\mathbf{k}} & -\Delta^*_{\mathbf{k}} \\ -\Delta_{\mathbf{k}} & -\epsilon_{\mathbf{k}} \end{bmatrix} \tag{4.29}$$

Next step is to perform the well-known Bogoliubov transform and reduce all the problem down to a simple ensemble of quantum pseudospins.

4.6.1 Bogoliubov transform and pseudospins picture

Let τ^α for $\alpha = x, y, z$ be the Pauli matrices. Define the pseudospin operator:

$$\hat{s}^\alpha_{\mathbf{k}} \equiv \hat{\Psi}^\dagger_{\mathbf{k}} \tau^\alpha \hat{\Psi}_{\mathbf{k}} \quad \text{for} \quad \alpha = x, y, z$$

As can be shown easily, these operators realize spin-1/2 algebra. $\hat{H}$ represents an ensemble of $L_x L_y$ independent spins subject to pseudo-magnetic fields. Note that, differently form App. **??** where the chemical potential is inserted later (because in Nambu formalism it accounts for a diagonal term) here the chemical potential is part of the z component of the pseudo-magnetic field, since

$$\begin{aligned} \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{-\mathbf{k}\downarrow} &= \hat{c}^\dagger_{\mathbf{k}\uparrow} \hat{c}_{\mathbf{k}\uparrow} + \hat{c}^\dagger_{-\mathbf{k}\downarrow} \hat{c}_{-\mathbf{k}\downarrow} \\ &= \hat{c}^\dagger_{\mathbf{k}\uparrow} \hat{c}_{\mathbf{k}\uparrow} - \hat{c}_{-\mathbf{k}\downarrow} \hat{c}^\dagger_{-\mathbf{k}\downarrow} + \mathbb{I} \\ &= \hat{\Psi}^\dagger_{\mathbf{k}} \tau^z \hat{\Psi}_{\mathbf{k}} + \mathbb{I} \end{aligned} \tag{4.30}$$

and then it follows

$$\begin{aligned} -\mu \hat{N} &= -\mu \sum_{\mathbf{k} \in \mathrm{BZ}} [\hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{-\mathbf{k}\downarrow}] \\ &= -\mu \sum_{\mathbf{k} \in \mathrm{BZ}} \hat{\Psi}^\dagger_{\mathbf{k}} \tau^z \hat{\Psi}_{\mathbf{k}} - \mu L_x L_y \end{aligned}$$

Then, adding a term $-\mu \hat{N}$ to $\hat{H}$, apart from an irrelevant total energy increase, changes the pseudo-field whose explicit form becomes

$$\mathbf{b}_{\mathbf{k}} \equiv \begin{bmatrix} -\operatorname{Re}\{\Delta_{\mathbf{k}}\} \\ -\operatorname{Im}\{\Delta_{\mathbf{k}}\} \\ \epsilon_{\mathbf{k}} - \mu \end{bmatrix} \tag{4.31}$$

This hamiltonian behaves as an ensemble of spins in local magnetic fields precisely as in Eq. (**??**),

$$\hat{H} - \mu \hat{N} = \sum_{\mathbf{k} \in \mathrm{BZ}} \mathbf{b}_{\mathbf{k}} \cdot \hat{\mathbf{s}}_{\mathbf{k}} \quad \text{where} \quad \hat{\mathbf{s}}_{\mathbf{k}} = \begin{bmatrix} \hat{s}^x_{\mathbf{k}} \\ \hat{s}^y_{\mathbf{k}} \\ \hat{s}^z_{\mathbf{k}} \end{bmatrix} \tag{4.32}$$

Proceed as in App. **??** and diagonalize via a rotation,

$$d_{\mathbf{k}} \equiv \begin{bmatrix} -E_{\mathbf{k}} & \\ & E_{\mathbf{k}} \end{bmatrix} \quad \text{being} \quad E_{\mathbf{k}} \equiv \sqrt{\xi^2_{\mathbf{k}} + |\Delta_{\mathbf{k}}|^2}$$

and $\xi_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} - \mu$. Given the pseudoangles

$$\tan(2\theta_{\mathbf{k}}) \equiv \frac{|\Delta_{\mathbf{k}}|}{\xi_{\mathbf{k}}} \qquad \tan(2\zeta_{\mathbf{k}}) \equiv \frac{\operatorname{Im}\{\Delta_{\mathbf{k}}\}}{\operatorname{Re}\{\Delta_{\mathbf{k}}\}}$$

the general diagonalizer will be an orthogonal rotation matrix

$$\begin{aligned} W_{\mathbf{k}} &= e^{i\left(\theta_{\mathbf{k}} - \frac{\pi}{2}\right)\tau^y} e^{i\zeta_{\mathbf{k}}\tau^z} \\ &= \begin{bmatrix} -\sin\theta_{\mathbf{k}} & -\cos\theta_{\mathbf{k}} \\ \cos\theta_{\mathbf{k}} & -\sin\theta_{\mathbf{k}} \end{bmatrix} \begin{bmatrix} e^{i\zeta_{\mathbf{k}}} & \\ & e^{-i\zeta_{\mathbf{k}}} \end{bmatrix} \\ &= \begin{bmatrix} -\sin\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} & -\cos\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} \\ \cos\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} & -\sin\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} \end{bmatrix} \end{aligned} \tag{4.33}$$

given by a rotation of angle $\zeta_{\mathbf{k}}$ around the z axis, to align the x axis with the field projection onto the xy plane, followed by a rotation around the y axis to anti-align with the pseudo-field.

4.6.2 BCS ground state properties

The MFT-BCS solution is given by a degenerate Fermi gas at ground state, whose quasi-particles occupy two bands $\pm E_{\mathbf{k}}$ and their fermionic operators are given by

$$\hat{\gamma}^{(-)}_{\mathbf{k}} \equiv \left[W_{\mathbf{k}}\hat{\Psi}_{\mathbf{k}}\right]_1 \qquad \hat{\gamma}^{(+)}_{\mathbf{k}} \equiv \left[W_{\mathbf{k}}\hat{\Psi}_{\mathbf{k}}\right]_2$$

The diagonalization operators are given by

$$\hat{\Gamma}_{\mathbf{k}} \equiv W_{\mathbf{k}}\hat{\Psi}_{\mathbf{k}} \quad \text{where} \quad \hat{\Gamma}_{\mathbf{k}} = \begin{bmatrix} \hat{\gamma}^{(-)}_{\mathbf{k}} \\ \hat{\gamma}^{(+)}_{\mathbf{k}} \end{bmatrix}$$

Using Eq. (**??**),

$$\left\langle [\hat{\Psi}^\dagger_{\mathbf{k}}]_i [\hat{\Psi}_{\mathbf{k}}]_j \right\rangle = [W_{\mathbf{k}}]_{1i}[W^\dagger_{\mathbf{k}}]_{j1} f\left(-E_{\mathbf{k}};\beta,0\right) + [W_{\mathbf{k}}]_{2i}[W^\dagger_{\mathbf{k}}]_{j2} f\left(E_{\mathbf{k}};\beta,0\right)$$

where in the Fermi-Dirac function chemical potential was set to zero, because it already was included in the diagonalized hamiltonian. Recalling Eq. (**??**), it follows for $i=1$, $j=2$

$$\langle \hat{\phi}^\dagger_{\mathbf{k}} \rangle = [W_{\mathbf{k}}]_{11}[W^\dagger_{\mathbf{k}}]_{21} f\left(-E_{\mathbf{k}};\beta,0\right) + [W_{\mathbf{k}}]_{21}[W^\dagger_{\mathbf{k}}]_{22} f\left(E_{\mathbf{k}};\beta,0\right) \tag{4.34}$$

$$= \frac{1}{2}\sin\left(2\theta_{\mathbf{k}}\right) e^{i2\zeta_{\mathbf{k}}} \tanh\left(\frac{\beta E_{\mathbf{k}}}{2}\right) \tag{4.35}$$

The last passage has been obtained by computing the matrix element from the explicit form of $W_{\mathbf{k}}$ of Eq. (4.33) and by the simple relation

$$\frac{1}{e^{-x}+1} - \frac{1}{e^{x}+1} = \frac{e^x-1}{e^x+1} = \tanh\left(\frac{x}{2}\right)$$

Similarly, for $i=1$, $j=1$

$$\langle \hat{c}^\dagger_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} \rangle = [W_{\mathbf{k}}]_{11}[W^\dagger_{\mathbf{k}}]_{11} f\left(-E_{\mathbf{k}};\beta,0\right) + [W_{\mathbf{k}}]_{21}[W^\dagger_{\mathbf{k}}]_{12} f\left(E_{\mathbf{k}};\beta,0\right) \tag{4.36}$$

$$= \sin^2\theta_{\mathbf{k}} f\left(-E_{\mathbf{k}};\beta,0\right) + \cos^2\theta_{\mathbf{k}} f\left(E_{\mathbf{k}};\beta,0\right)$$

$$= \sin^2\theta_{\mathbf{k}} \tanh\left(\frac{\beta E_{\mathbf{k}}}{2}\right) + f\left(E_{\mathbf{k}};\beta,0\right) \tag{4.37}$$

and for $i=2$, $j=2$

$$\langle \hat{c}_{-\mathbf{k}\downarrow}\hat{c}^\dagger_{-\mathbf{k}\downarrow} \rangle = [W_{\mathbf{k}}]_{12}[W^\dagger_{\mathbf{k}}]_{21} f\left(-E_{\mathbf{k}};\beta,0\right) + [W_{\mathbf{k}}]_{22}[W^\dagger_{\mathbf{k}}]_{22} f\left(E_{\mathbf{k}};\beta,0\right) \tag{4.38}$$

$$= \cos^2\theta_{\mathbf{k}} f\left(-E_{\mathbf{k}};\beta,0\right) + \sin^2\theta_{\mathbf{k}} f\left(E_{\mathbf{k}};\beta,0\right)$$

$$= \cos^2\theta_{\mathbf{k}} \tanh\left(\frac{\beta E_{\mathbf{k}}}{2}\right) + f\left(E_{\mathbf{k}};\beta,0\right) \tag{4.39}$$

Gap function. Eqns. (4.34), (4.35) give us both the algorithmic formula (first row) and its theoretical counterpart (second row) to compute the order parameters in the HF approach at each point in k-space (k_x, k_y). We can finally derive the BCS self-consistency equation

$$\Delta_{\mathbf{k}} \equiv \frac{1}{2}\sum_{\mathbf{k}'}\left[V^{(s)} + V_{\mathbf{k}\mathbf{k}'}\right] \frac{|\Delta_{\mathbf{k}'}|}{\sqrt{\xi^2_{\mathbf{k}'} + |\Delta_{\mathbf{k}'}|^2}} \exp\left\{ i \arctan\left(\frac{\mathrm{Im}\{\Delta_{\mathbf{k}'}\}}{\mathrm{Re}\{\Delta_{\mathbf{k}'}\}}\right)\right\} \tanh\left(\frac{\beta}{2}\sqrt{\xi^2_{\mathbf{k}'} + |\Delta_{\mathbf{k}'}|^2}\right)$$

$$= \frac{1}{2}\sum_{\mathbf{k}'}\left[V^{(s)} + V_{\mathbf{k}\mathbf{k}'}\right] \frac{\Delta_{\mathbf{k}'}}{\sqrt{\xi^2_{\mathbf{k}'} + |\Delta_{\mathbf{k}'}|^2}} \tanh\left(\frac{\beta}{2}\sqrt{\xi^2_{\mathbf{k}'} + |\Delta_{\mathbf{k}'}|^2}\right) \tag{4.40}$$

The whole point of the HF algorithm is to find an iterative solution for each symmetry channel, using the self-consistency equation projection of Tab. 4.3.

System density. The z component of the spin operators is related to density: using Eq. (**??**),

$$\langle \hat{\Psi}_{\mathbf{k}}^{\dagger} \tau^z \hat{\Psi}_{\mathbf{k}} \rangle = \left\langle [\hat{\Psi}_{\mathbf{k}}^{\dagger}]_1 [\hat{\Psi}_{\mathbf{k}}]_1 \right\rangle - \left\langle [\hat{\Psi}_{\mathbf{k}}^{\dagger}]_2 [\hat{\Psi}_{\mathbf{k}}]_2 \right\rangle$$

Proceed as previously, and from Eq. (4.30),

$$\begin{aligned}
\langle \hat{n}_{\mathbf{k}\uparrow} \rangle + \langle \hat{n}_{-\mathbf{k}\downarrow} \rangle &= 1 + \langle \hat{\Psi}_{\mathbf{k}}^{\dagger} \tau^z \hat{\Psi}_{\mathbf{k}} \rangle \\
&= 1 + \left(|[W_{\mathbf{k}}]_{11}|^2 - |[W_{\mathbf{k}}]_{12}|^2 \right) f\left(-E_{\mathbf{k}}; \beta, 0\right) \\
&\quad + \left(|[W_{\mathbf{k}}]_{21}|^2 - |[W_{\mathbf{k}}]_{22}|^2 \right) f\left(E_{\mathbf{k}}; \beta, 0\right) && (4.41) \\
&= 1 - \cos\left(2\theta_{\mathbf{k}}\right) \tanh\left(\frac{\beta E_{\mathbf{k}}}{2}\right) && (4.42)
\end{aligned}$$

coherently with Eqns. (4.37) and (4.39). The expectation value for the density is needed in order to extract the optimal chemical potential μ for the target density we aim to simulate at the given parametrization. This is numerically obtained by using Eq. (4.41) directly on the diagonalization matrix of $h_{\mathbf{k}}$.

Ground state at zero temperature. Let $\beta \to +\infty$. The basic BCS grond state is well known and identical in the present discussion. It is easily obtained by considering the zero-temperature ground state of the pseudospin hamiltonian of Eq. (4.29),

$$|\mathrm{BCS}\rangle = \bigotimes_{\mathbf{k}} |2\theta_{\mathbf{k}}\rangle \tag{4.43}$$

where $|2\theta_{\mathbf{k}}\rangle$ is the up-state for the $\mathbf{k}$-th rotated pseudospin operator $\hat{\Gamma}_{\mathbf{k}}^{\dagger} \tau^z \hat{\Gamma}_{\mathbf{k}}$ (consider a diagram analogous to Fig. 3.1). Said state can be expressed, making use of the non-rotated pseudospin operator $\hat{\Psi}_{\mathbf{k}}^{\dagger} \tau^z \hat{\Psi}_{\mathbf{k}}$ up and down states

$$\text{up: } |\Uparrow_{\mathbf{k}}\rangle \qquad \text{down: } |\Downarrow_{\mathbf{k}}\rangle$$

by the means of a simple Bloch's representation [Redo this computation...],

$$|2\theta_{\mathbf{k}}\rangle = \cos\theta_{\mathbf{k}} |\Downarrow_{\mathbf{k}}\rangle - \sin\theta_{\mathbf{k}} |\Uparrow_{\mathbf{k}}\rangle \tag{4.44}$$

Due to Eq. (4.30) it holds

$$\begin{aligned}
\left[\hat{\Psi}_{\mathbf{k}}^{\dagger} \tau^z \hat{\Psi}_{\mathbf{k}} + \mathbb{I}\right] |\Uparrow_{\mathbf{k}}\rangle = 2 |\Uparrow_{\mathbf{k}}\rangle \quad &\Longrightarrow \quad |\Uparrow_{\mathbf{k}}\rangle = -\hat{\phi}_{\mathbf{k}}^{\dagger} |\Omega_{\mathbf{k}}\rangle \\
\left[\hat{\Psi}_{\mathbf{k}}^{\dagger} \tau^z \hat{\Psi}_{\mathbf{k}} + \mathbb{I}\right] |\Downarrow_{\mathbf{k}}\rangle = 0 \quad &\Longrightarrow \quad |\Downarrow_{\mathbf{k}}\rangle = |\Omega_{\mathbf{k}}\rangle
\end{aligned}$$

being $|\Omega_{\mathbf{k}}\rangle$ the local electron vacuum,

$$|\Omega\rangle = \bigotimes_{\mathbf{k}} |\Omega_{\mathbf{k}}\rangle$$

Taking Eq. (4.43), combined with Eq. (4.44), we get

$$\begin{aligned}
|\mathrm{BCS}\rangle &= \bigotimes_{\mathbf{k}} \left[\cos\theta_{\mathbf{k}} |\Downarrow_{\mathbf{k}}\rangle - \sin\theta_{\mathbf{k}} |\Uparrow_{\mathbf{k}}\rangle\right] \\
&= \bigotimes_{\mathbf{k}} \left[\cos\theta_{\mathbf{k}} + \sin\theta_{\mathbf{k}} \hat{\phi}_{\mathbf{k}}^{\dagger}\right] |\Omega_{\mathbf{k}}\rangle && (4.45)
\end{aligned}$$

which is the well-known coherent BCS state.

4.6.3 Renormalization of the bare bands

We now deal with the renormalization of bare bands anticipated in Sec. 4.5.2. Taking together Eqns. (4.27), (4.34) and (4.35) we get:

$$V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}\rangle\hat{c}^{\dagger}_{j\sigma}\hat{c}_{i\sigma}+\text{h.c.}\right]=\frac{2V}{L_xL_y}\sum_{\mathbf{q},\mathbf{q}'}\left[\cos\left(\delta q_x\right)+\cos\left(\delta q_y\right)\right]$$
$$\times\left[\left(\sin^2\theta_{\mathbf{q}}\tanh\left(\frac{\beta E_{\mathbf{q}}}{2}\right)+f\left(E_{\mathbf{q}};\beta,0\right)\right)\hat{c}^{\dagger}_{\mathbf{q}'\uparrow}\hat{c}_{\mathbf{q}'\uparrow}\right.$$
$$\left.+\left(1-\cos^2\theta_{-\mathbf{q}}\tanh\left(\frac{\beta E_{-\mathbf{q}}}{2}\right)-f\left(E_{-\mathbf{q}};\beta,0\right)\right)\hat{c}^{\dagger}_{\mathbf{q}'\downarrow}\hat{c}_{\mathbf{q}'\downarrow}\right] \tag{4.46}$$

Spatial inversion symmetry guarantees

$$\cos^2\theta_{-\mathbf{q}}=\cos^2\theta_{\mathbf{q}}\quad\text{as well as}\quad E_{-\mathbf{q}}=E_{\mathbf{q}}$$

By simple algebraic manipulation is easy to see:

$$1-\cos^2\theta_{\mathbf{q}}\tanh\left(\frac{\beta E_{\mathbf{q}}}{2}\right)-f\left(E_{\mathbf{q}};\beta,0\right)=\sin^2\theta_{\mathbf{q}}\tanh\left(\frac{\beta E_{\mathbf{q}}}{2}\right)+1-\tanh\left(\frac{\beta E_{\mathbf{q}}}{2}\right)-f\left(E_{\mathbf{q}};\beta,0\right)$$
$$=\sin^2\theta_{\mathbf{q}}\tanh\left(\frac{\beta E_{\mathbf{q}}}{2}\right)+f\left(E_{\mathbf{q}};\beta,0\right)$$

Since

$$1-\tanh\left(\frac{x}{2}\right)-\frac{1}{e^x+1}=\frac{1}{e^x+1}$$

Use now:

$$\sin^2\theta_{\mathbf{q}}=\frac{1-\cos(2\theta_{\mathbf{q}})}{2}$$
$$=\frac{1}{2}-\frac{\xi_{\mathbf{q}}}{2\sqrt{\xi^2_{\mathbf{q}}+\left|\Delta_{\mathbf{q}}\right|^2}}$$

and let:

$$g_{\mathbf{q}}\equiv\left[\frac{1}{2}-\frac{\xi_{\mathbf{q}}}{2\sqrt{\xi^2_{\mathbf{q}}+\left|\Delta_{\mathbf{q}}\right|^2}}\right]\tanh\left(\frac{\beta}{2}\sqrt{\xi^2_{\mathbf{q}}+\left|\Delta_{\mathbf{q}}\right|^2}\right)+f\left(\sqrt{\xi^2_{\mathbf{q}}+\left|\Delta_{\mathbf{q}}\right|^2};\beta,0\right)$$

which further simplifies to

$$g_{\mathbf{q}}\equiv\frac{1}{2}-\frac{\xi_{\mathbf{q}}}{2\sqrt{\xi^2_{\mathbf{q}}+\left|\Delta_{\mathbf{q}}\right|^2}}\tanh\left(\frac{\beta}{2}\sqrt{\xi^2_{\mathbf{q}}+\left|\Delta_{\mathbf{q}}\right|^2}\right) \tag{4.47}$$

Thus, the original o.s. Fock term retrieved from Wick's theorem on the EHM reduces to

$$V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}\rangle\hat{c}^{\dagger}_{j\sigma}\hat{c}_{i\sigma}+\text{h.c.}\right]=\frac{2V}{L_xL_y}\sum_{\mathbf{q},\mathbf{q}'}\sum_{\sigma}\left[\cos\left(\delta q_x\right)+\cos\left(\delta q_y\right)\right]g_{\mathbf{q}}\hat{c}^{\dagger}_{\mathbf{q}'\sigma}\hat{c}_{\mathbf{q}'\sigma}$$

Recall now the result of Eq. (1.20),

$$\cos\left(\delta q_x\right)+\cos\left(\delta q_y\right)=\frac{1}{2}\sum_{\gamma}\varphi^{(\gamma)}_{\mathbf{q}}\varphi^{(\gamma)*}_{\mathbf{q}'}\quad\text{for }\gamma\in\{s^*,p_x,p_y,d_{x^2-y^2}\}$$

which gives (sum over γ is intended within the aforementioned symmetries)

$$\frac{2V}{L_xL_y}\sum_{\mathbf{q},\mathbf{q}'}\sum_{\sigma}\left[\cos\left(\delta q_x\right)+\cos\left(\delta q_y\right)\right]g_{\mathbf{q}}\hat{c}^{\dagger}_{\mathbf{q}'\sigma}\hat{c}_{\mathbf{q}'\sigma}=\frac{V}{L_xL_y}\sum_{\mathbf{q},\mathbf{q}'}\sum_{\sigma,\gamma}\varphi^{(\gamma)}_{\mathbf{q}}\varphi^{(\gamma)*}_{\mathbf{q}'}g_{\mathbf{q}}\hat{c}^{\dagger}_{\mathbf{q}'\sigma}\hat{c}_{\mathbf{q}'\sigma} \tag{4.48}$$

Evidently from Eq. (4.47), $g_{\mathbf{q}}$ is an even function of the momentum, a feature that ensures

$$\frac{1}{L_xL_y}\sum_{\mathbf{q}}\varphi^{(p_\ell)}_{\mathbf{q}}g_{\mathbf{q}}=0\quad\text{for }\ell\in\{x,y\}$$

General result. Let us now divide the discussion in two parts. First, let us deal with the theoretical general result of the above calculations. Define the s^* and $d_{x^2-y^2}$ components as

$$g^{(s^*)} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{q}} (\cos q_x + \cos q_y) g_{\mathbf{q}} \qquad g^{(d)} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{q}} (\cos q_x - \cos q_y) g_{\mathbf{q}}$$

and from Eq. (4.48) we get the final form

$$\frac{2V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_{\sigma} \left[\cos(\delta q_x) + \cos(\delta q_y)\right] g_{\mathbf{q}} \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma}$$
$$= V \sum_{\mathbf{q}\sigma} \left[g^{(s^*)} (\cos q_x + \cos q_y) + g^{(d)} (\cos q_x - \cos q_y) \right] \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \quad (4.49)$$

This equation reduces fully the initial o.s. Fock component of $\hat{H}_V$ to a simpler, decoupled form. Here, similarly to what we observed for AF phase in Sec. **??**, the bare bands are renormalized,

$$\tilde{\epsilon}_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} + V \left[g^{(s^*)} (\cos k_x + \cos k_y) + g^{(d)} (\cos k_x - \cos k_y) \right] \quad (4.50)$$

As is done in next paragraph, the s^* is actually a rigid t renormalization, which however requires a little care.

Pure x-wave superconductivity. As long as we work in a precise sector ($s \oplus s^*$ as well as $d_{x^2-y^2}$) with a gap function completely contained in said symmetry sector, we can rule out the $d_{x^2-y^2}$-wave component of $g_{\mathbf{q}}$ from Eq. (4.50)

$$\frac{1}{L_x L_y} \sum_{\mathbf{q}} \varphi^{(d)}_{\mathbf{q}} g_{\mathbf{q}} = 0$$

The result above is valid only if $\Delta_{\mathbf{q}}$ is described only by a single symmetry harmonics. This is because under a rotation of angle $\pi/2$ the two functions behave differently:

$$(q_x, q_y) \to (-q_y, q_x) \quad : \quad \begin{cases} \varphi^{(d)}_{\mathbf{q}} \to -\varphi^{(d)}_{\mathbf{q}} \\ g_{\mathbf{q}} \to g_{\mathbf{q}} \end{cases}$$

The only remainder of Eq. (4.50) is its s^* part

$$\tilde{\epsilon}_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} + V g^{(s^*)} (\cos k_x + \cos k_y) \quad (4.51)$$

and since $\epsilon_{\mathbf{k}} = -2t(\cos k_x + \cos k_y)$ we obtain an equation essentially identical to the one expressing the hopping renormalization in the AF phase, Eq. (**??**). Indeed, let $w \equiv g^{(s^*)}/2$ be the new HF parameter, we get

$$\tilde{t} \equiv t - wV \quad (4.52)$$

Once again the hopping parameter shift is rigid and controlled directly by V. As done for the AF phase, we will denote the new bands as:

$$\xi_{\mathbf{k}} \to \tilde{\xi}_{\mathbf{k}} \qquad E_{\mathbf{k}} \to \tilde{E}_{\mathbf{k}}$$

4.7 SU free energy density

BCS essentially relies on the insurgence of an effective free fermionic field, described by the Bogoliubov bands, and the condensation of Cooper pairs. Superconducting state is favored by the lowering of free energy granted by the shifting of much of the spectral weight to a lower-energy maximum [**zhou1992density**]. "Condensation energy" is simply the energy difference between

the superconducting state and the normal state, and its negativity indicates instability towards superconducting order. It's pretty straightforward to derive the ground state energy for the standard BCS state. When performing Wick's contractions,

$$\begin{aligned}\langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta \rangle \sim \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle \hat{c}_\gamma \hat{c}_\delta + \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \langle \hat{c}_\gamma \hat{c}_\delta \rangle - \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle \langle \hat{c}_\gamma \hat{c}_\delta \rangle \quad & \text{Cooper terms} \\ - \langle \hat{c}^\dagger_\alpha \hat{c}_\gamma \rangle \hat{c}^\dagger_\beta \hat{c}_\delta - \hat{c}^\dagger_\alpha \hat{c}_\gamma \langle \hat{c}^\dagger_\beta \hat{c}_\delta \rangle + \langle \hat{c}^\dagger_\alpha \hat{c}_\gamma \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\delta \rangle \quad & \text{Fock terms} \\ + \langle \hat{c}^\dagger_\alpha \hat{c}_\delta \rangle \hat{c}^\dagger_\beta \hat{c}_\gamma + \hat{c}^\dagger_\alpha \hat{c}_\delta \langle \hat{c}^\dagger_\beta \hat{c}_\gamma \rangle - \langle \hat{c}^\dagger_\alpha \hat{c}_\delta \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\gamma \rangle \quad & \text{Hartree terms}\end{aligned}$$

Form each of the three rows above, the last term is the "leftover" part which shifts energy. Thus it can be ignored in derivation, but becomes important when considering the energy in the end. The EHM hamiltonian can be written as:

$$\begin{aligned}\hat{H} = \sum_{\mathbf{k}} \xi_{\mathbf{k}} \left[\hat{c}^\dagger_{\mathbf{k}\uparrow} \hat{c}_{\mathbf{k}\uparrow} + \mathbb{I} - \hat{c}_{-\mathbf{k}\downarrow} \hat{c}^\dagger_{-\mathbf{k}\downarrow} \right] - \sum_{\mathbf{k},\mathbf{k}'} \left[V^{(s)} + V_{\mathbf{k}\mathbf{k}'} \right] \hat{\phi}^\dagger_{\mathbf{k}} \hat{\phi}_{\mathbf{k}'} \\ - \frac{2V}{L_x L_y} \sum_{\mathbf{k},\mathbf{k}'} \sum_{\sigma} \left[\cos(\delta k_x) + \cos(\delta k_y) \right] \hat{n}_{\mathbf{k}\sigma} \hat{n}_{\mathbf{k}'\sigma}\end{aligned} \tag{4.53}$$

as in Eq. (4.25). Following the scheme above, the "leftovers" of the MFT hamiltonian are just

$$\sum_{\mathbf{k},\mathbf{k}'} \Delta_{\mathbf{k}} \langle \hat{\phi}_{\mathbf{k}'} \rangle - \frac{2V}{L_x L_y} \sum_{\mathbf{k},\mathbf{k}'} \sum_{\sigma} \left[\cos(\delta k_x) + \cos(\delta k_y) \right] g_{\mathbf{k}} g_{\mathbf{k}'}$$

Further developing this result and applying Eqns. (4.35), (1.20), these last terms can be written as:

$$\sum_{\mathbf{k}} \frac{|\Delta_{\mathbf{k}}|^2}{2\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2} \right) - 2V L_x L_y \sum_{\gamma} \left| g^{(\gamma)} \right|^2 \qquad \gamma \in \{s^*, d_{x^2-y^2}\} \tag{4.54}$$

as for Eq. (4.49). Notice the presence of a volume factor in the $\mathcal{O}\left(g^2\right)$ correction.

Now, we retrieve the expression for the MFT free energy density [**hutchinson2021mixed**] simply by adding four contributions:

1. Bogoliubov free quasiparticles thermal ground state energy density,

$$\begin{aligned}\frac{1}{L_x L_y} \sum_{\mathbf{k}} \tilde{E}_{\mathbf{k}} \left[f(\tilde{E}_{\mathbf{k}}; \beta, 0) - f(-\tilde{E}_{\mathbf{k}}; \beta, 0) \right] &= \frac{1}{L_x L_y} \sum_{\mathbf{k}} \tilde{E}_{\mathbf{k}} \left[f(\tilde{E}_{\mathbf{k}}; \beta, 0) - \left(1 - f(\tilde{E}_{\mathbf{k}}; \beta, 0) \right) \right] \\ &= \frac{2}{L_x L_y} \sum_{\mathbf{k}} \tilde{E}_{\mathbf{k}} f(\tilde{E}_{\mathbf{k}}; \beta, 0) - \frac{1}{L_x L_y} \sum_{\mathbf{k}} \tilde{E}_{\mathbf{k}}\end{aligned}$$

   having used the simple property of the Fermi-Dirac distribution $f(-x; \beta, 0) = 1 - f(-x; \beta, 0)$.

2. The term originated by the identity produced by fermionic anticommutation rules in Eq. (4.53),

$$\frac{1}{L_x L_y} \sum_{\mathbf{k}} \tilde{\xi}_{\mathbf{k}}$$

   averaged on the lattice.

3. The "leftovers" of Eq. (4.54), divided by $L_x L_y$.

4. The entropic free energy

$$-Ts$$

   with $T = 1/k_B \beta$ and s the free Fermi gas entropy density for the Bogoliubov quasiparticles,

$$s = -\frac{2k_B}{L_x L_y} \sum_{\mathbf{k}} \left[\left(1 - f(\tilde{E}_{\mathbf{k}}; \beta, 0) \right) \ln \left(1 - f(\tilde{E}_{\mathbf{k}}; \beta, 0) \right) + f(\tilde{E}_{\mathbf{k}}; \beta, 0) \ln f(\tilde{E}_{\mathbf{k}}; \beta, 0) \right]$$

We use the identity:

$$\ln\left(1-\frac{1}{e^x+1}\right)=x+\ln\left(\frac{1}{e^x+1}\right)$$

and reduce the above expression to:

$$s=-\frac{2k_B}{L_xL_y}\sum_{\mathbf{k}}\Big[\ln\Big(1-f(\tilde{E}_{\mathbf{k}};\beta,0)\Big)+f(\tilde{E}_{\mathbf{k}};\beta,0)\ln f(\tilde{E}_{\mathbf{k}};\beta,0)$$
$$-\beta\tilde{E}_{\mathbf{k}}f(\tilde{E}_{\mathbf{k}};\beta,0)-f(\tilde{E}_{\mathbf{k}};\beta,0)\ln f(\tilde{E}_{\mathbf{k}};\beta,0)\Big]$$

Second and fourth terms cancel out, while the third compensates the first one of contribution 1.

All together, the MFT averaged free energy looks like this:

$$f_{\mathrm{MFT}}[\Delta_{\mathbf{k}},g_{\mathbf{k}}]=\frac{1}{L_xL_y}\sum_{\mathbf{k}}\left[\tilde{\xi}_{\mathbf{k}}-\tilde{E}_{\mathbf{k}}+\frac{|\Delta_{\mathbf{k}}|^2}{2\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)\right]$$
$$-2V\sum_{\gamma\in\{s^*,d\}}\left|g^{(\gamma)}\right|^2+\frac{2}{\beta L_xL_y}\sum_{\mathbf{k}}\ln\Big(1-f(\tilde{E}_{\mathbf{k}};\beta,0)\Big) \quad (4.55)$$

The two main differences with the commonly known expressions for f_{MFT} [**hutchinson2021mixed**, **senarath2025mixed**] are the presence of bands renormalization , the absence of $-\mu n$ [Why?], and the $\mathcal{O}\left(g^2\right)$ correction. The latter, not only is negative for attractive V, a feature that can lead to superconducting phase energy lowering and stabilisation, but also it depends explicitly on the spectral amplitude of the s^* and d-wave components of $g_{\mathbf{k}}$. Since as seen the d-wave part of $g_{\mathbf{k}}$ arises for mixed symmetries, this details seems to indicate an energetic preference for multi-symmetry phases.

4.8 Numerical simulations

[To be continued...]

4.8.1 HF iterative algorithm sketch

[To be continued...]